Module II

Semiconductor Devices and Applications

Introduction to semiconductors: conductors, insulators and semiconductors, P-type and N-type semiconductors,
PN junction diode: construction, working principle and V-I characteristics, Diode as a switch,
Rectifiers: Half-wave and Full-wave/Bridge – working and applications, Zener diode as a voltage regulator.

Structure of Atom

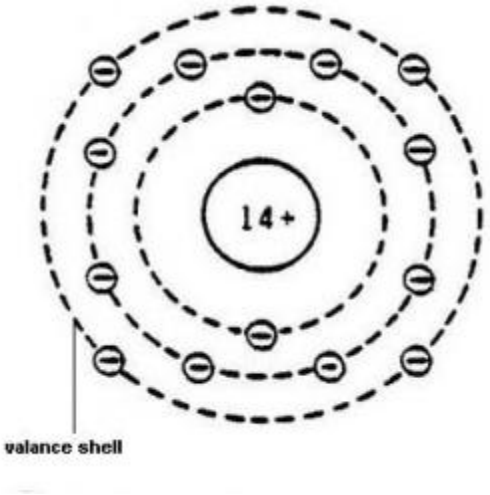
- **Atom** :
  - Each element is made up of atom
  - An atom possesses the unique characteristics of that element.

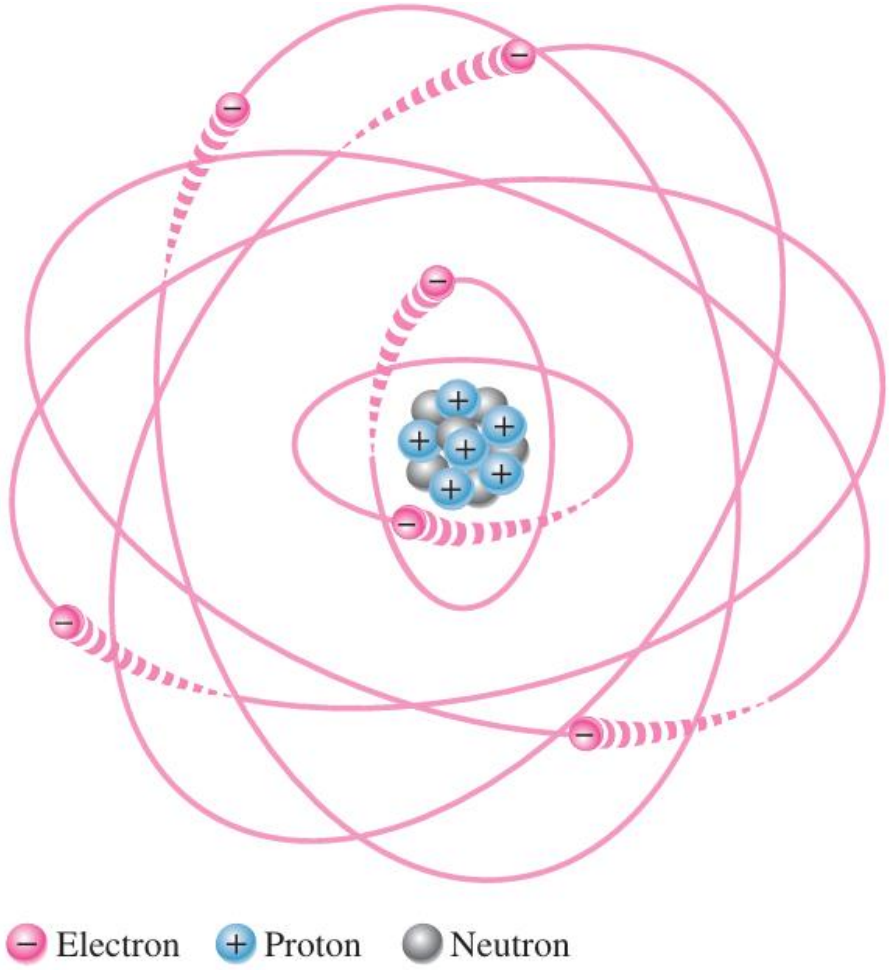
- **Structure of an Atom :**

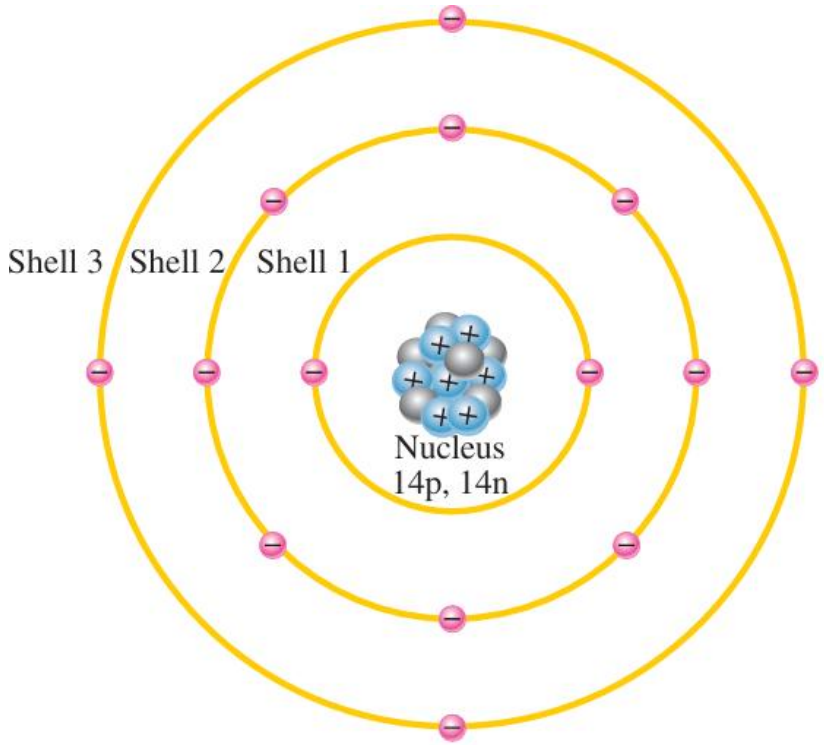
1.Centrally located : Nucleus
  - Protons (Positively charged)
  - Neutrons (Neutral)

2.Orbit :
  - Electrons (Negatively charged)

3.Outermost shell :
  - Valence electrons loosely bound to nucleus




Electron
Proton
Neutron



The Maximum Number of Electrons in Each Shell: The maximum number of electrons (Ne) that can exist in each shell of an atom is a fact of nature and can be calculated by the formula

$$N_e = 2n^2$$

Conductivity of Materials

- Conductivity of a material can be defined as the ability of a material to carry current.

- This can be determined with the help of valence electrons.

Conductors, Insulators and Semiconductors

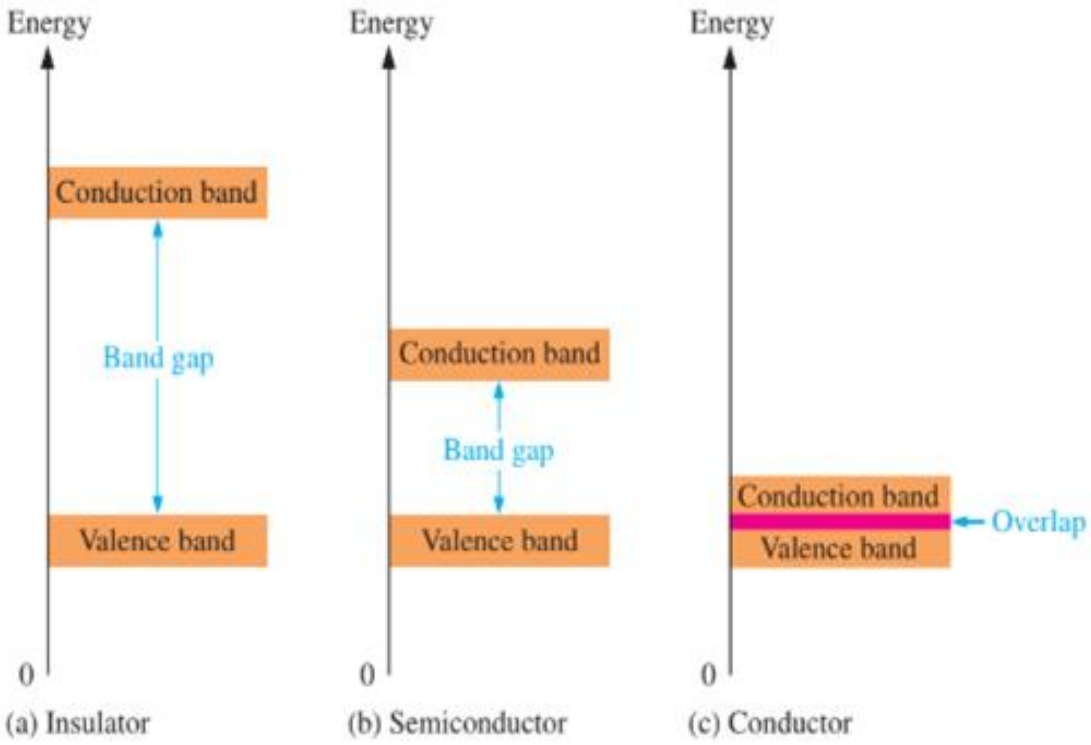
- **Conductors** –A material through which current flows easily is called a conductor.
  - Eg.Copper ,Aluminium ,silver etc.

- **Insulators** - A material through which current cannot flow is called a Insulator.
  - Eg.Wood,Rubber ,Glass

- **Semiconductor** -A semiconductor is a material that is between conductors and insulators in its ability to conduct electrical current.
  - Eg. Silicon and Germanium

Energy Bands of each Material

An energy band is a range of energy levels that electrons are allowed to occupy inside a solid material.



◀ FIGURE 1–7

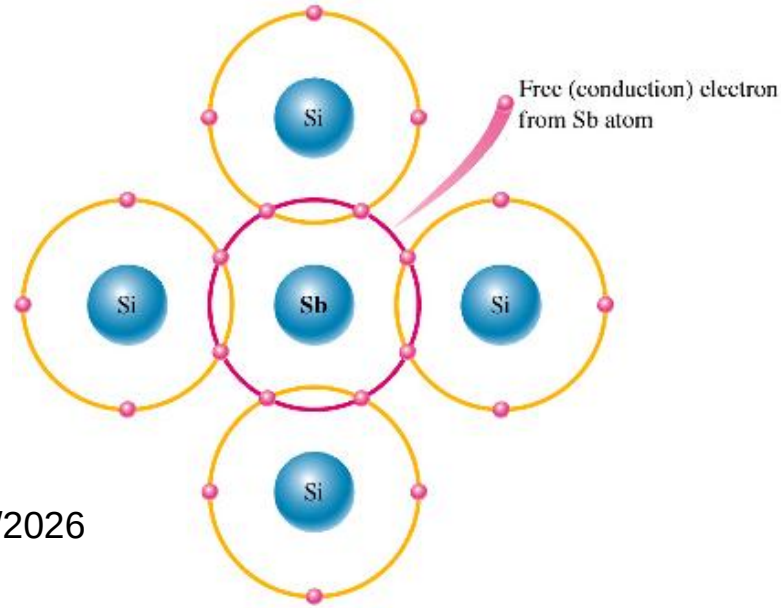
Energy diagrams for the three types of materials.

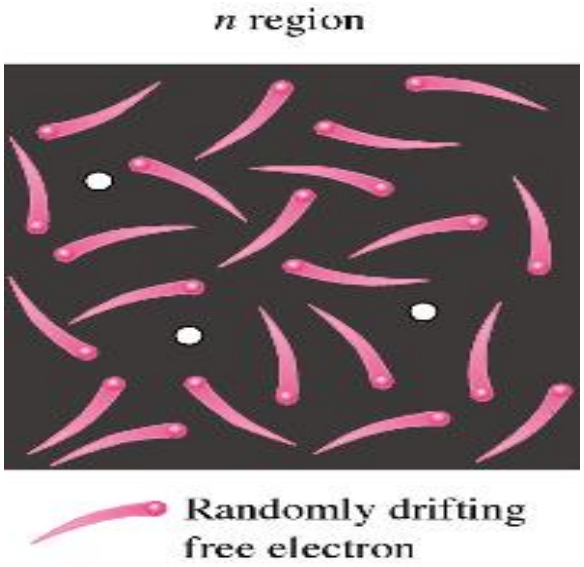
Introduction to Semiconductors

- A semiconductor in its pure form is called Intrinsic semiconductor.
- However with proper chemical process the semiconductor can be converted to a conductor .
- The conductivity of a semiconductor can be increased if we add another material with the semiconductor
- The process of adding impurity/ another material with a semiconductor to increase its conductivity is called as **Doping**.
- The doped semiconductor is called Extrinsic semiconductor.
- Two types of semiconductors after obtained after doping
  1. N-Type
  2. P-type

Formation of N-type Semiconductors,

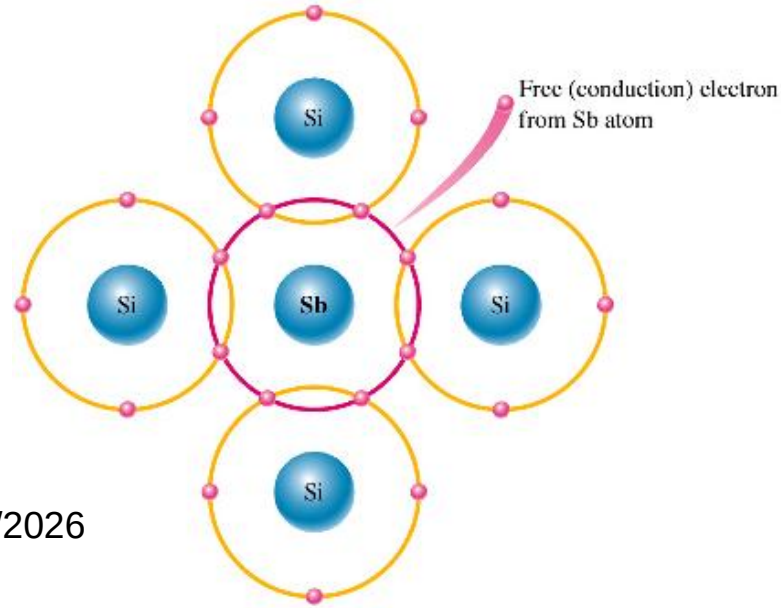
- To increase the number of conduction-band electrons in intrinsic semiconductor ,pentavalent impurity atoms are added.
- These are atoms with five valence electrons such as arsenic (As), phosphorus (P), bismuth (Bi), and antimony (Sb).
- Each pentavalent atom forms covalent bonds with four adjacent silicon atoms.
- Four of the antimony atom's valence electrons are used to form the covalent bonds with silicon atoms, leaving one extra electron.

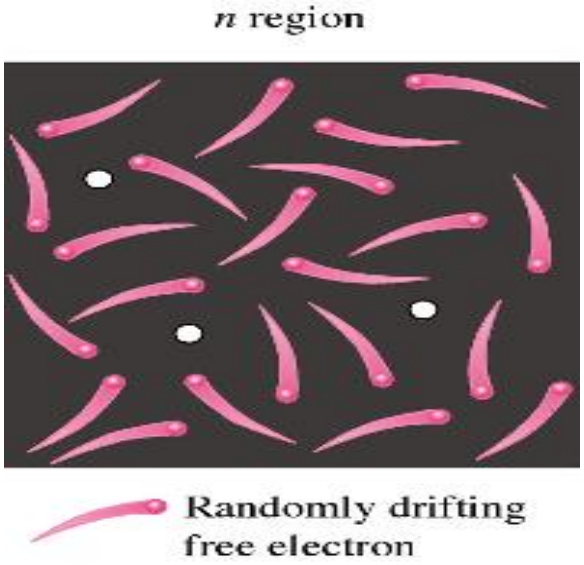




Formation of N-type Semiconductors,

- This extra electron becomes a conduction electron because it is not involved in bonding.
- Because the pentavalent atom gives up an electron, it is often called a donor atom.
- The electrons are called the majority carriers in n-type material.
- Holes in an n-type material are called minority carriers.

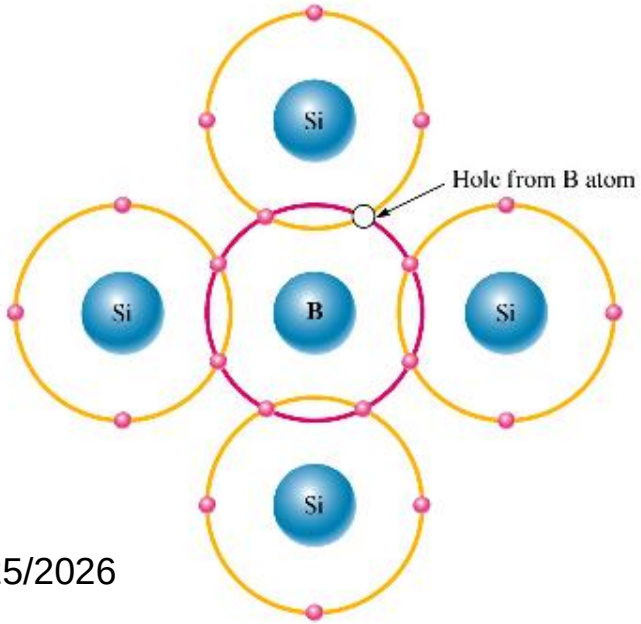


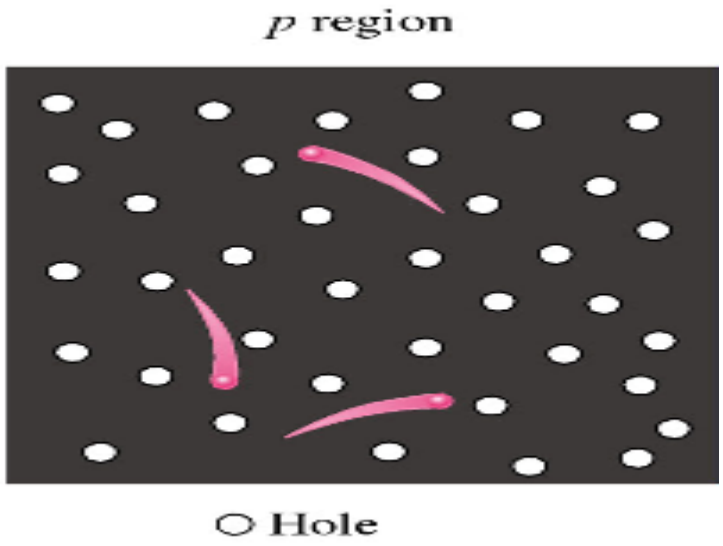


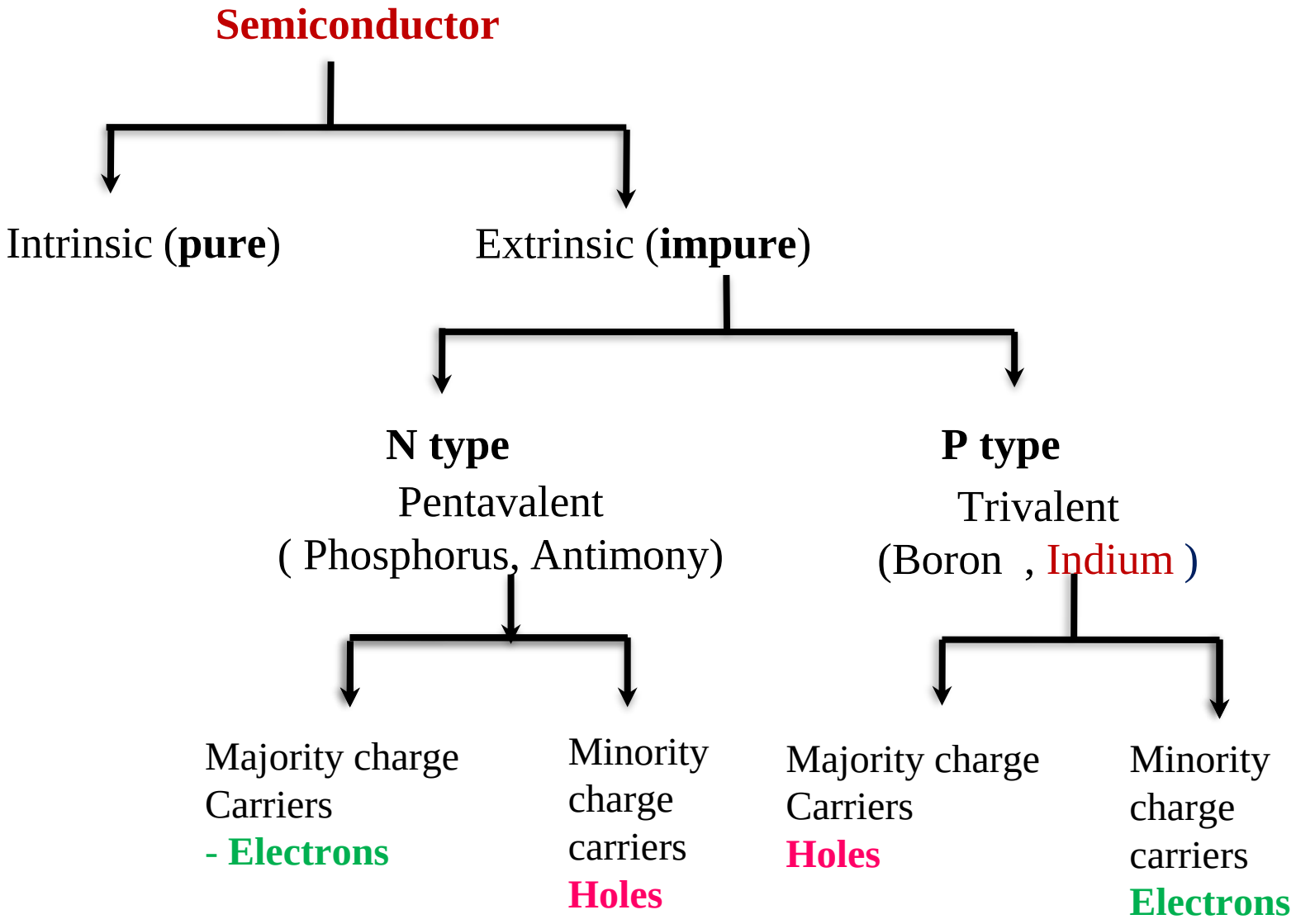
Formation of P-type Semiconductors,

- A P-type material consist of silicon atoms and trivalent atoms such as boron.
- The boron atom adds a hole when it bounds with the silicon atoms.

- However since the number of protons and number of electrons are equal throughout the material, there is no net charge in the material and so it is neutral.






Semiconductor
Intrinsic (pure)
Extrinsic (impure)
N type
Pentavalent
(Phosphorus, Antimony)
P type
Trivalent
(Boron , Indium)
Majority charge Carriers
- Electrons
Minority charge carriers
Holes
Majority charge Carriers
Holes
Minority charge carriers
Electrons

P-N junction Diode

- The P-N junction itself forms the most basic semiconductor device called semiconductor diode.
- The meaning of “diode” is the device having “two electrodes”.
- As shown in figure the diode has two electrodes one each for the two regions on either sides of the junction.



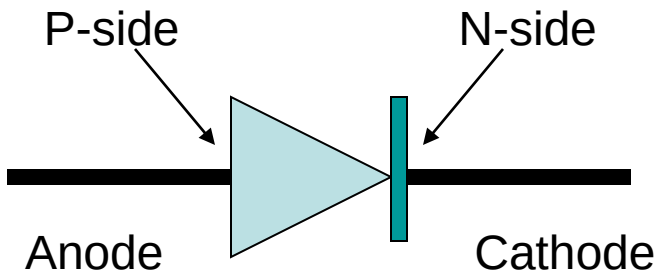
P-N junction forms a semiconductor diode

Symbol of P-N junction Diode

- The arrowhead symbol points in the direction of conventional current through the device.
- This current will flow through the diode if and only if an external voltage source is connected to it with appropriate polarities.



P-N junction forms a semiconductor diode



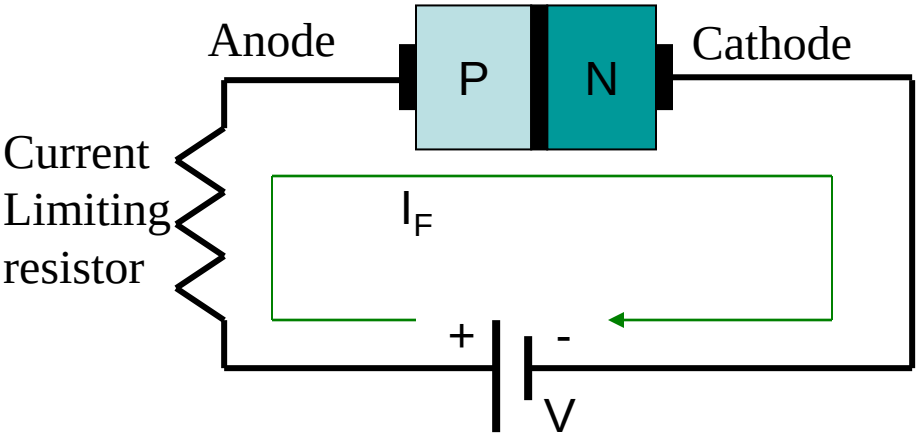
Circuit symbol of a diode

Biasing of P-N junction Diode

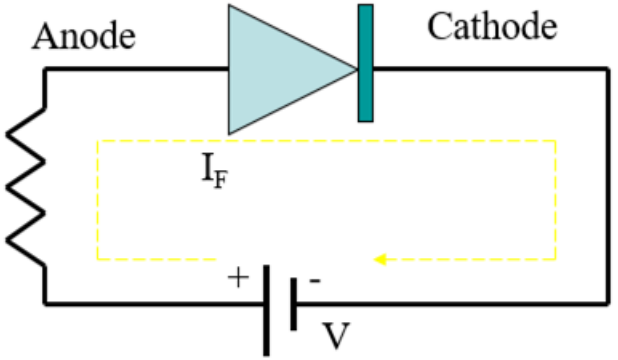
- The process of applying external DC voltage to the semiconductor diode is called Biasing .

- The biasing can be of two types:
  1. Forward bias,
  2. Reverse bias.

Forward biasing of a p-n junction Diode

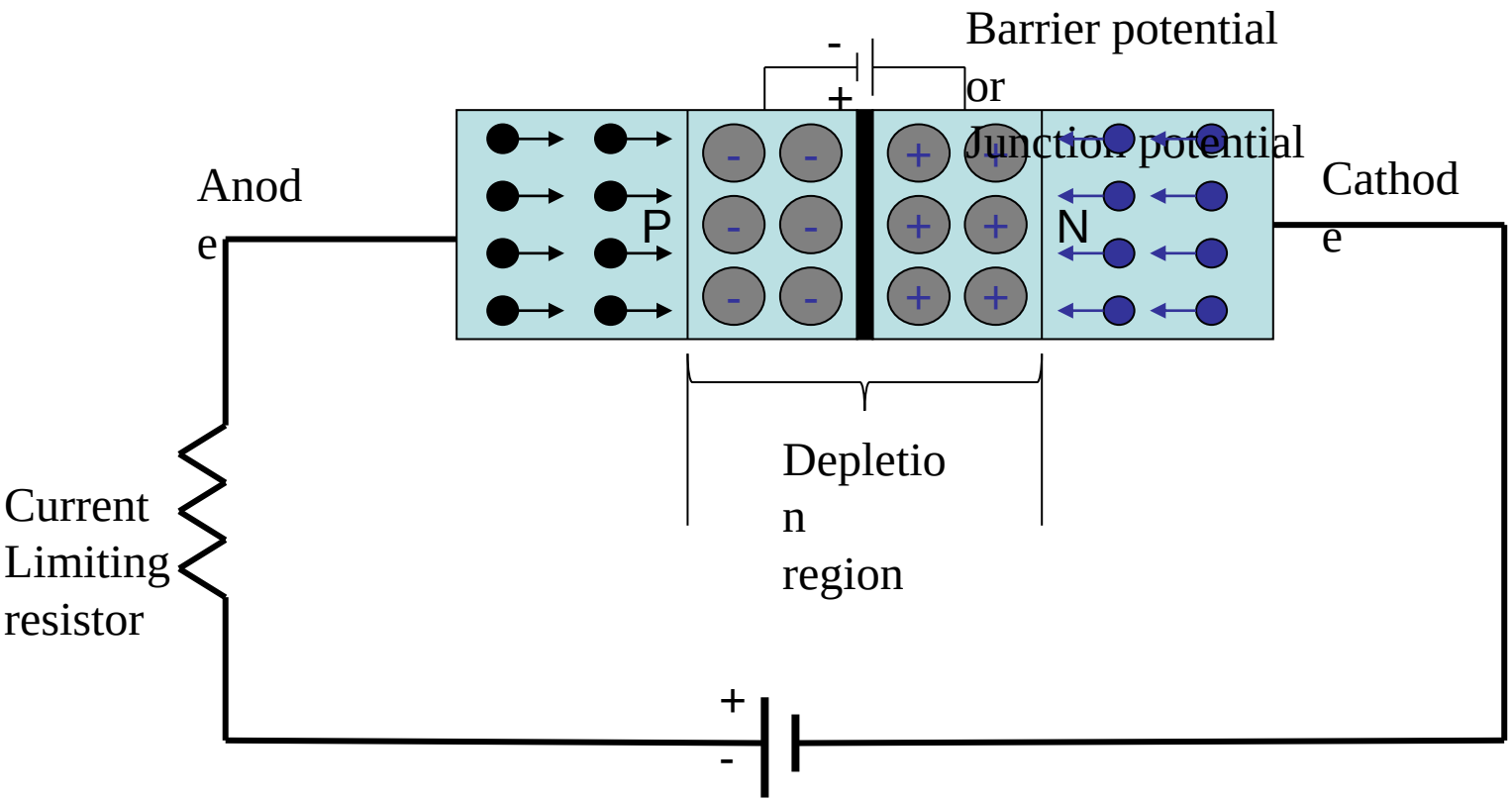


Forward biasing of diode



Symbolic representation of a forward biased diode

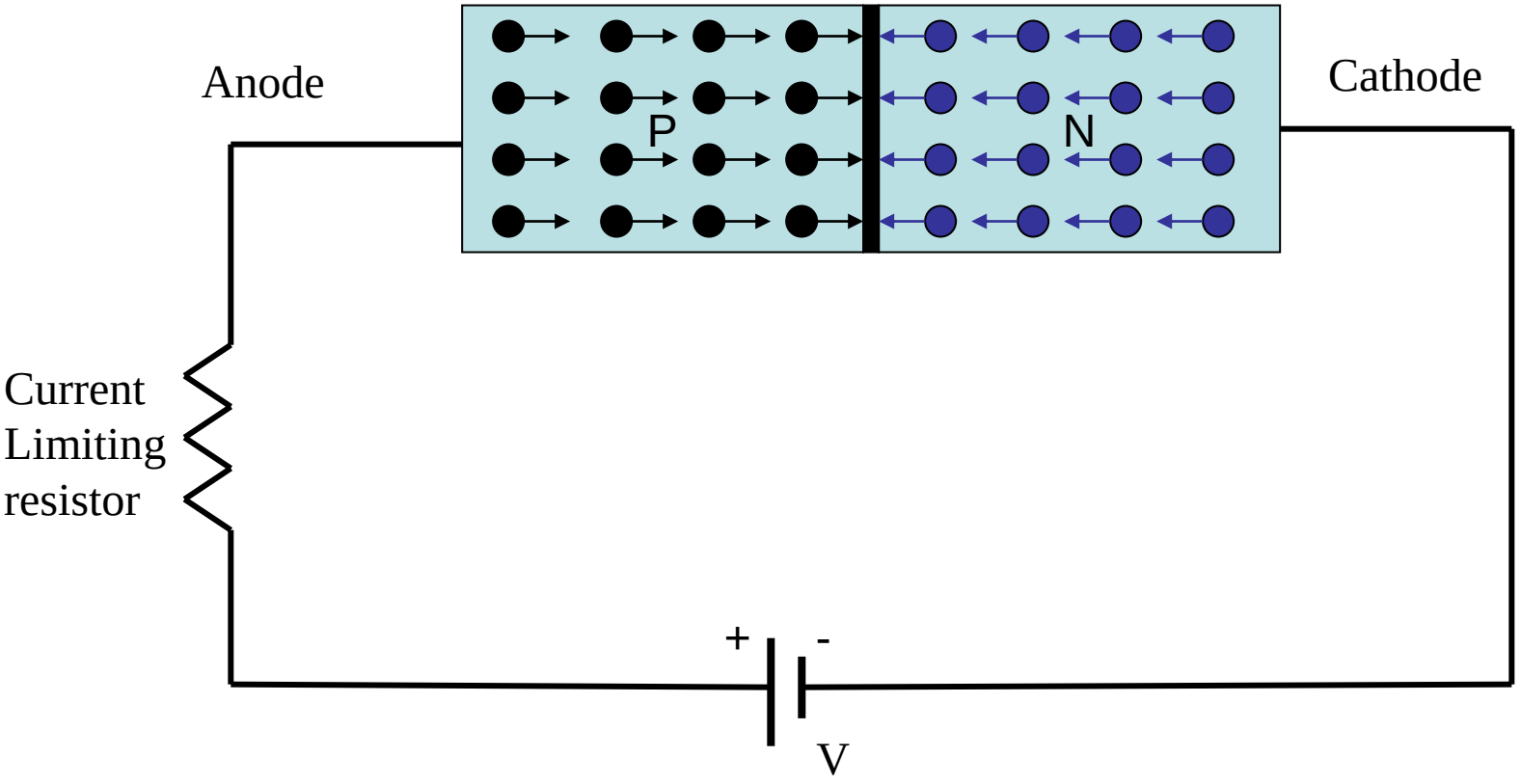
Operation of Forward biased Diode



- Due to the negative terminal of external source connected to the n-region, free electrons ,from n-side are pushed towards the p-side.

- Similarly the positive end of the supply will push holes , from p-side towards the n-side.

Operation of Forward biased Diode



- Due to reduction in depletion region width the barrier potential will also reduce.
- At a particular value of V the depletion region will collapse. Now there is no opposition to the flow of electrons and holes.

Forward characteristics of a diode

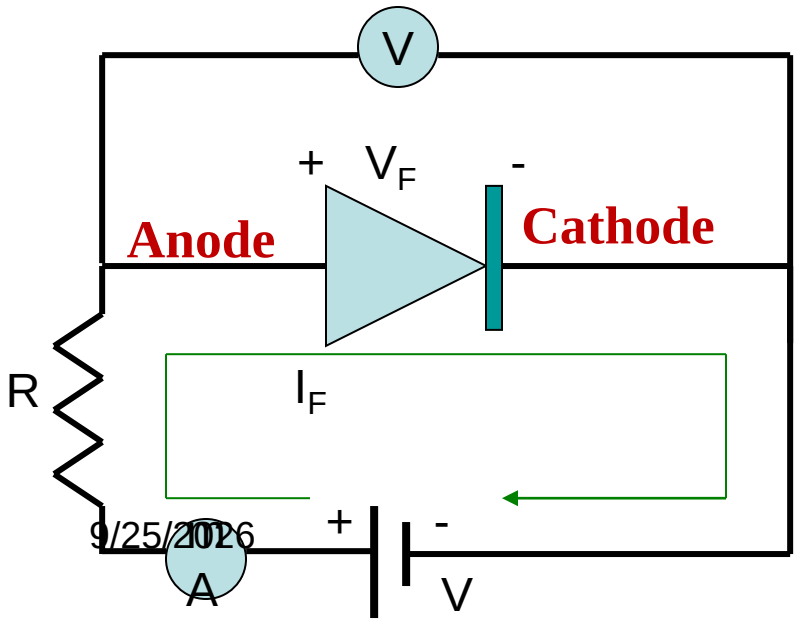
- The behavior of a p-n junction diode is indicated by its volt-ampere characteristics (V-I Characteristics).

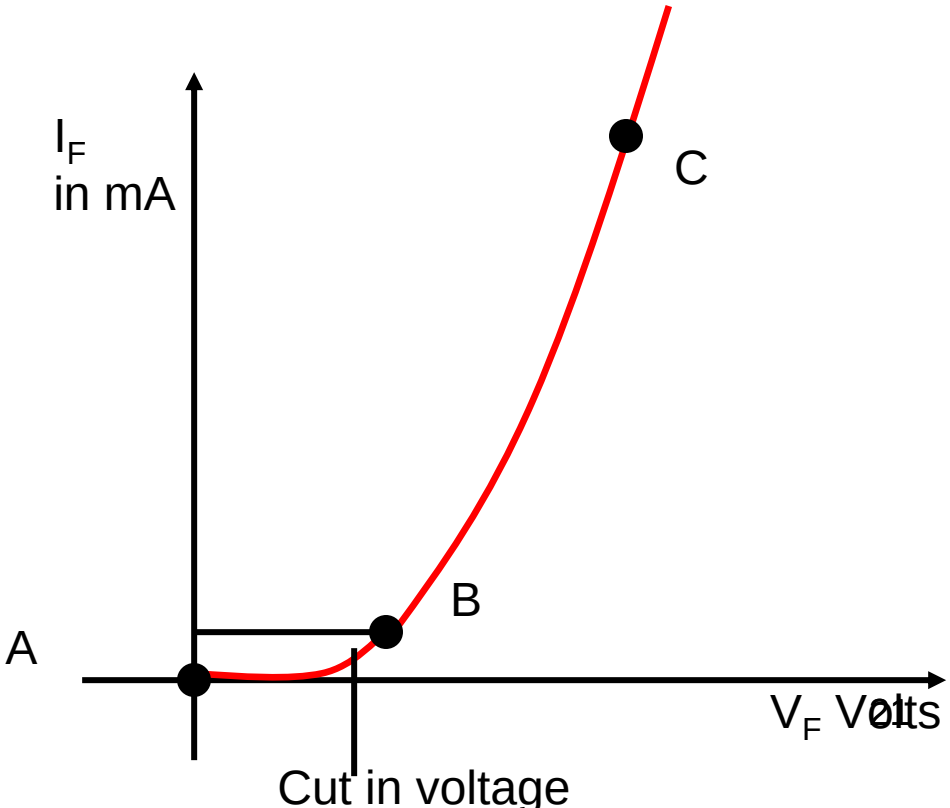
- The V-I characteristics of a p-n junction diode is a graph of voltage across the diode versus the current flowing through it.

- The V-I characteristics can be divided into two parts.
  1. Forward characteristics.
  2. Reverse Characteristics.

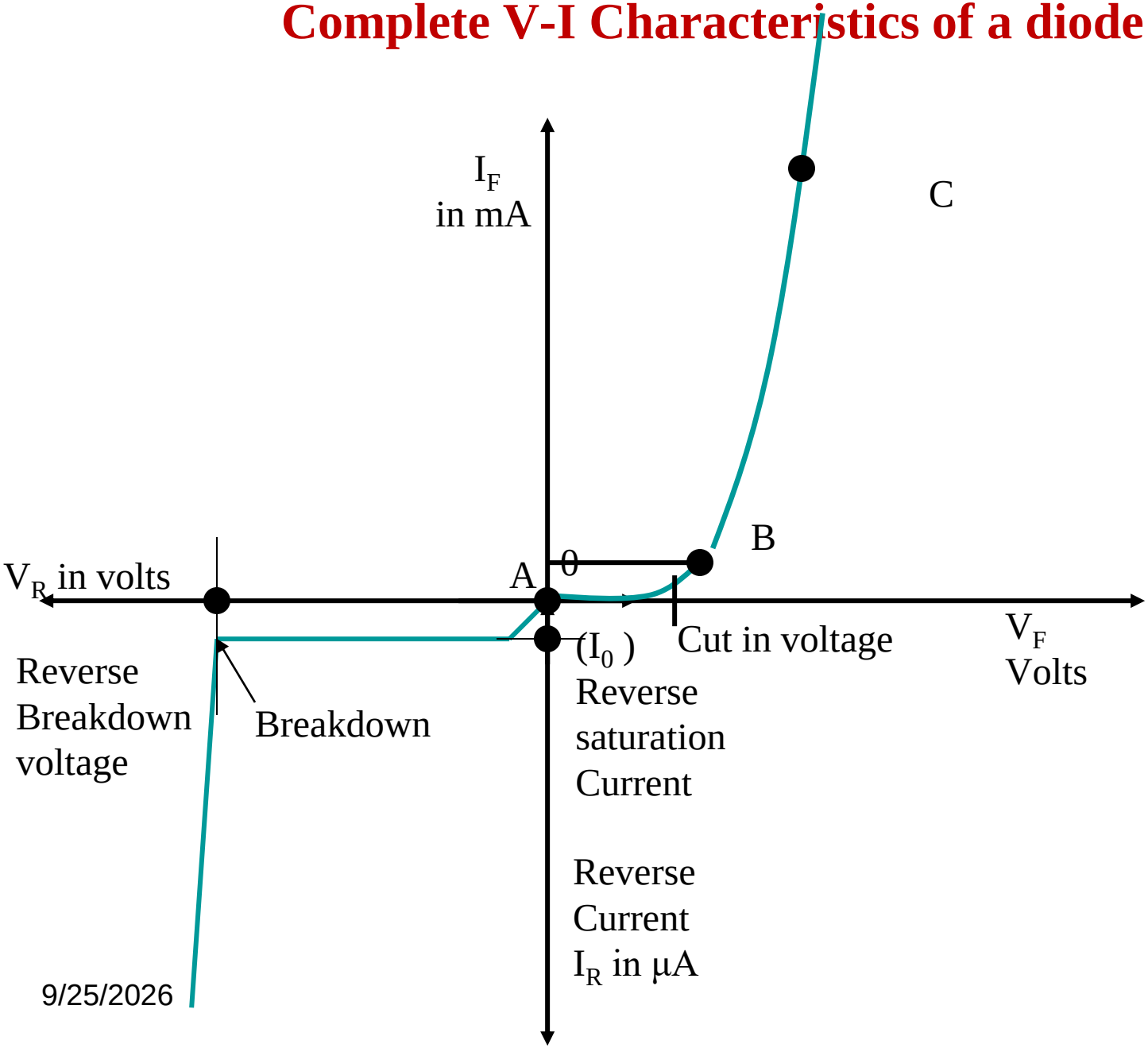
Forward Characteristics Of A Diode

- The forward characteristics is divided into two portions, AB and BC as shown in figure.

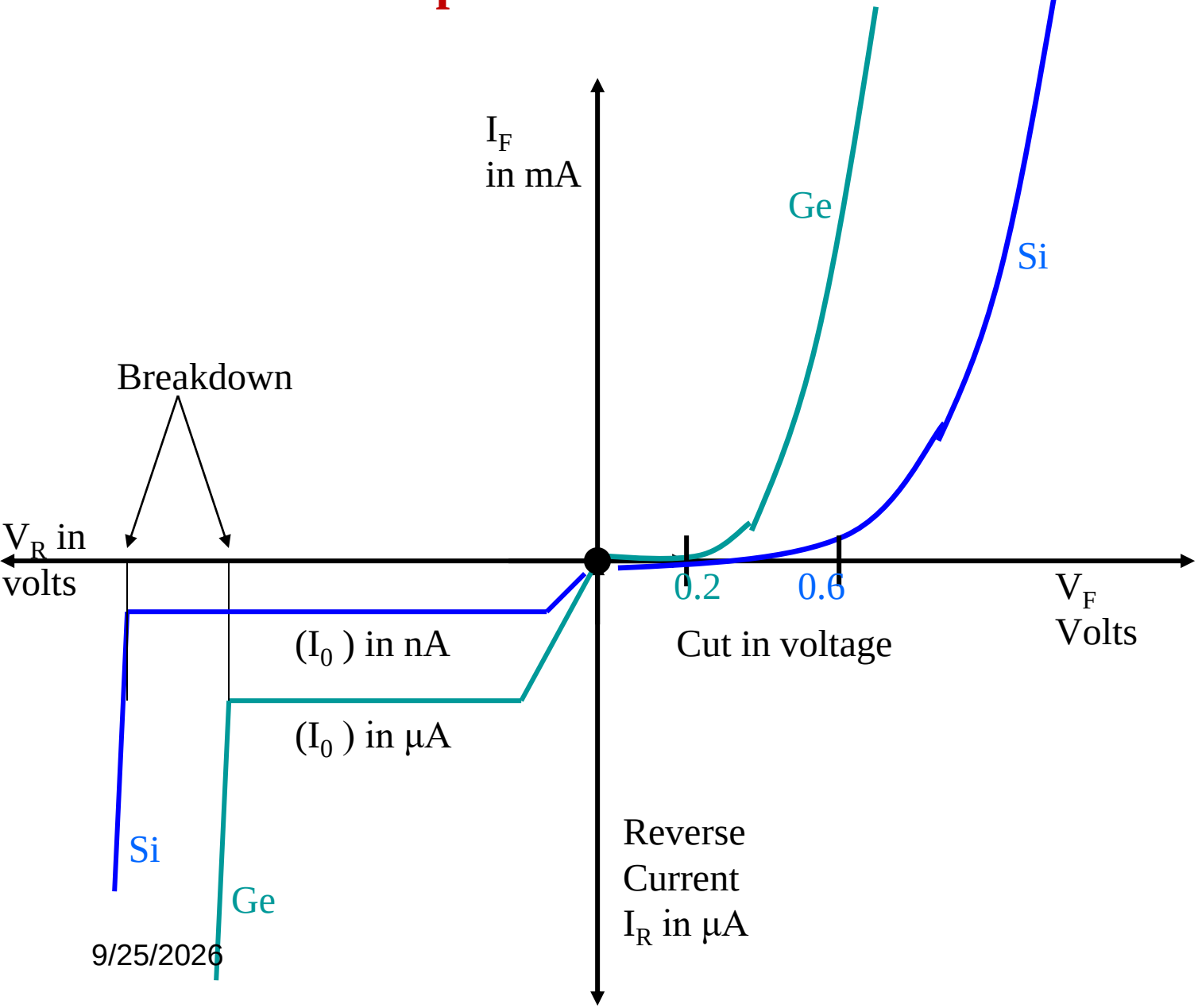




Complete V-I Characteristics of a diode



Complete V-I Characteristics of a diode



Breakdown Due To The Avalanche Effect

When a very large reverse voltage is applied to a diode the events take place in the following sequence :

- Due to large reverse voltage the velocity of the minority carriers will increase to a great extent. Therefore the kinetic energy with them will also increase.
- While traveling, these minority carriers will collide with the stationary atoms and impart some of the kinetic energy to the valence electrons present in the covalent bonds.
- Due to additionally acquired energy, these valence electrons will break the covalent bonds and jump into the conduction band to become free for conduction.
- Now these free electrons will be accelerated and they knock out some more valence electrons by means of collisions. This “chain reaction” is called as “Avalanche effect”.
- In a very short time, a large number of free minority electrons will be available for conduction and a large reverse current will flow through the reverse biased diode. The Avalanche breakdown does taken place.

Why Should Avoid Breakdown

- At the time of the avalanche breakdown, a large reverse voltage appears across the diode and a large reverse current is flowing through it.

- Therefore a large power is dissipated in the diode.

- The junction temperature of the diode may exceed its safe limits and the diode will be damaged permanently. Therefore the reverse breakdown should always be avoided.

Breakdown Due To The Zener Effect

The reverse breakdown can take place due to another effect called Zener effect.

- Due to heavy doping of p and n-sides of the diode, the depletion region is narrow in the reverse biased condition. All the reverse voltage V appears across the depletion region.

- Therefore the electric field which is the voltage per unit distance is very intense across the depletion region.

- This intense electric field can pull some of he valence electrons by breaking the covalent bonds. These electrons then become free electrons.

- A large number of such electrons can constitute a large reverse current through the diode. This is called as the breakdown due to zener effect.

Mathematical Expression For The Diode Current

1. The diode current I_D has an exponential shape. It is expressed as:

$$I_D = I_O [e^{V / \eta V_T} - 1]$$

Where Io = reverse saturation current

V_T = T/11,600 and it is called volt equivalent of temperature

η = 1 for germanium
= 2 for silicon

T = temperature in ^{O}K

V = Voltage across the diode

Mathematical Expression For The Forward Current (I_F)

2. The diode current I_F has an exponential shape. It is expressed as:

$$I_F = I_O [e^{V_F / \eta V_T} - 1]$$

if $e^{V_F / \eta V_F} >> 1$ then

$$I_F = I_O e^{V_F / \eta V_T}$$

this expression shows that the forward current is of exponential nature

Mathematical Expression For The Reverse Current (I_R)

substitute $I_D = I_R$ and $V = -V_R$ in equation we get

$$\mathbf{I_R = I_O [e^{-V_R / \eta V_T} - 1]}$$

but $e^{-VR / \eta Vt} << 1$ then

$$\mathbf{I_R = - I_O}$$

this expression shows that the reverse current I_R of diode is negative and constant , equals to Io (reverse saturation current)

Example

- The forward current through a silicon diode is 10 mA at room temperature (27^0 C) . The corresponding forward voltage is 0.75 V. calculate the reverse saturation current Io .

Given IF = 10 mA

V = 0.75 V

T = 27^0C = 300^0K

η = 2 for a silicon

V_T = T/ 11600

= 26 mV

$$I_F = I_O [e^{Vf / \eta Vt} - 1]$$

Io = ?

Example

- The forward current through a silicon diode is 10 mA at room temperature (27^0 C) . The corresponding forward voltage is 0.75 V. calculate the reverse saturation current Io .

Given IF = 10 mA

V = 0.75 V

T = 27^0C = 300^0K

$\eta = 2$ for a silicon

V_T = T/ 11600

= 26 mV

$$I_F = I_O [e^{Vf / \eta Vt} - 1]$$

Io = 5.446 nA

Applications Of P-N Junction Diode

Some of the application of a P-N junction diode are as follows :

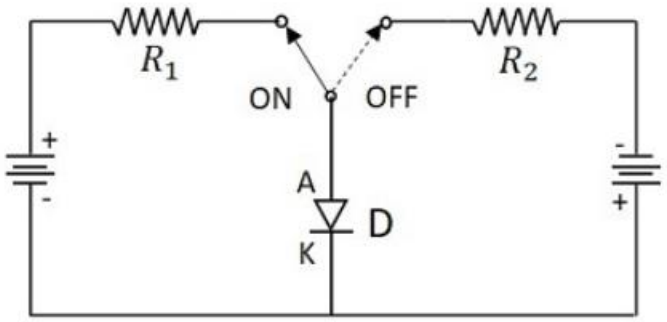
- Rectifier circuits
- Clipping and clamping of circuit
- Voltage multipliers
- A. M. detection
- Feedback diodes
- Freewheeling diodes
- etc……

Working of Diode as a Switch

Whenever a specified voltage is exceeded, the diode resistance gets increased, making the diode reverse biased and it acts as an open switch.

Whenever the voltage applied is below the reference voltage, the diode resistance gets decreased, making the diode forward biased, and it acts as a closed switch.

The following circuit explains the diode acting as a switch.



A switching diode has a PN junction in which P-region is lightly doped and N-region is heavily doped.

The above circuit symbolizes that the diode gets ON when positive voltage forward biased the diode and it gets OFF when negative voltage reverse biased the diode.

Working of Diode as a Switch

While changing the bias conditions, the diode undergoes a **transient response**. The response of a system to any sudden change from an equilibrium position is called as transient response.

The sudden change from forward to reverse and from reverse to forward bias, affects the circuit.

The time taken to respond to such sudden changes are classified as below.

The time taken before the diode recovers its steady state is called as **Recovery Time**.

The time interval taken by the diode to switch from reverse biased state to forward biased state is called as **Forward Recovery Time.(**tfr**)**

The time interval taken by the diode to switch from forward biased state to reverse biased state is called as **Reverse Recovery Time. (**tfr**)**

- **Review**

1. What is a pn junction?
2. Explain what diffusion is.
3. Describe the depletion region.
4. Explain what the barrier potential is and how it is created ?
5. What is the typical value of the barrier potential for a Silicon diode ?
6. What is the typical value of the barrier potential for a Germanium diode ?

Breakdown

- As the reverse voltage is increased, the reverse saturation current remains constant equals to Io if the temperature is constant.

- But as the reverse voltage reaches the breakdown voltage value, a large current flows through the diode.

- Operation in the breakdown region should be avoided because the diode may be damaged due to excessive power dissipation.

- Typically the reverse breakdown voltage for a p-n junction diode is in the range of 50 to 100 V.

Zener Diode

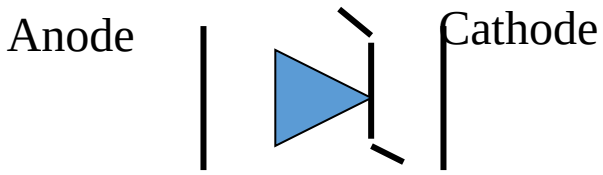
- "zener diode" is a special type p-n junction semiconductor diode. Is construction is similar to that of conventional p-n junction diode.

- However in constructing the zener diodes, the reverse breakdown voltage is adjusted precisely between 3V to 200V.

- The doping level of the impurity added to manufacture the zener diodes is controlled in order to adjust the precise value of breakdown voltage.

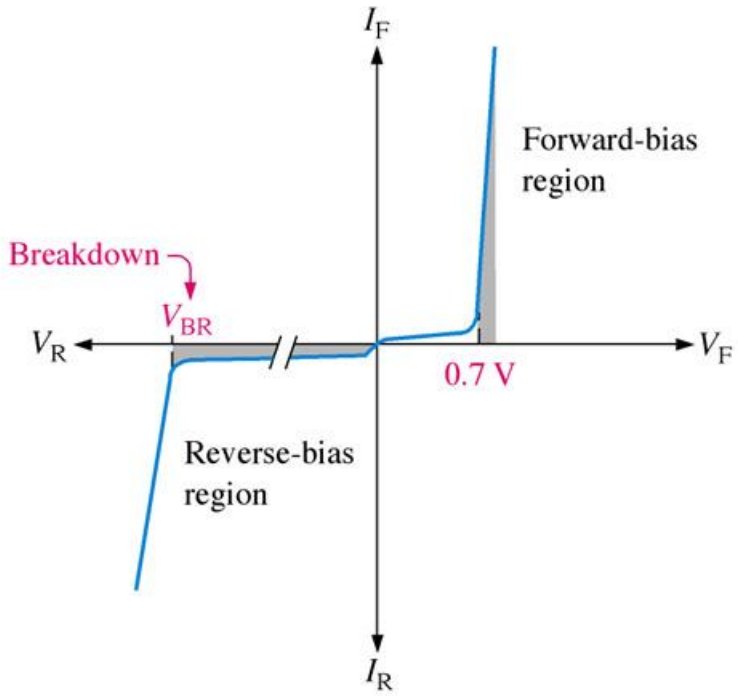
Circuit Symbol And Biasing Of A Zener Diode

- The circuit symbol of a zener diode is as shown in figure.

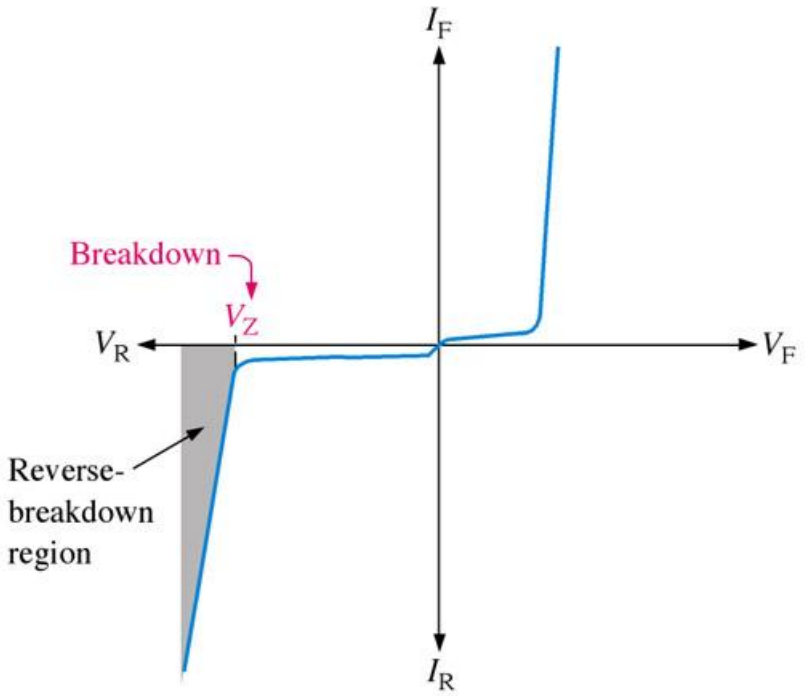


Circuit symbol

General Diode V-I Characteristic.

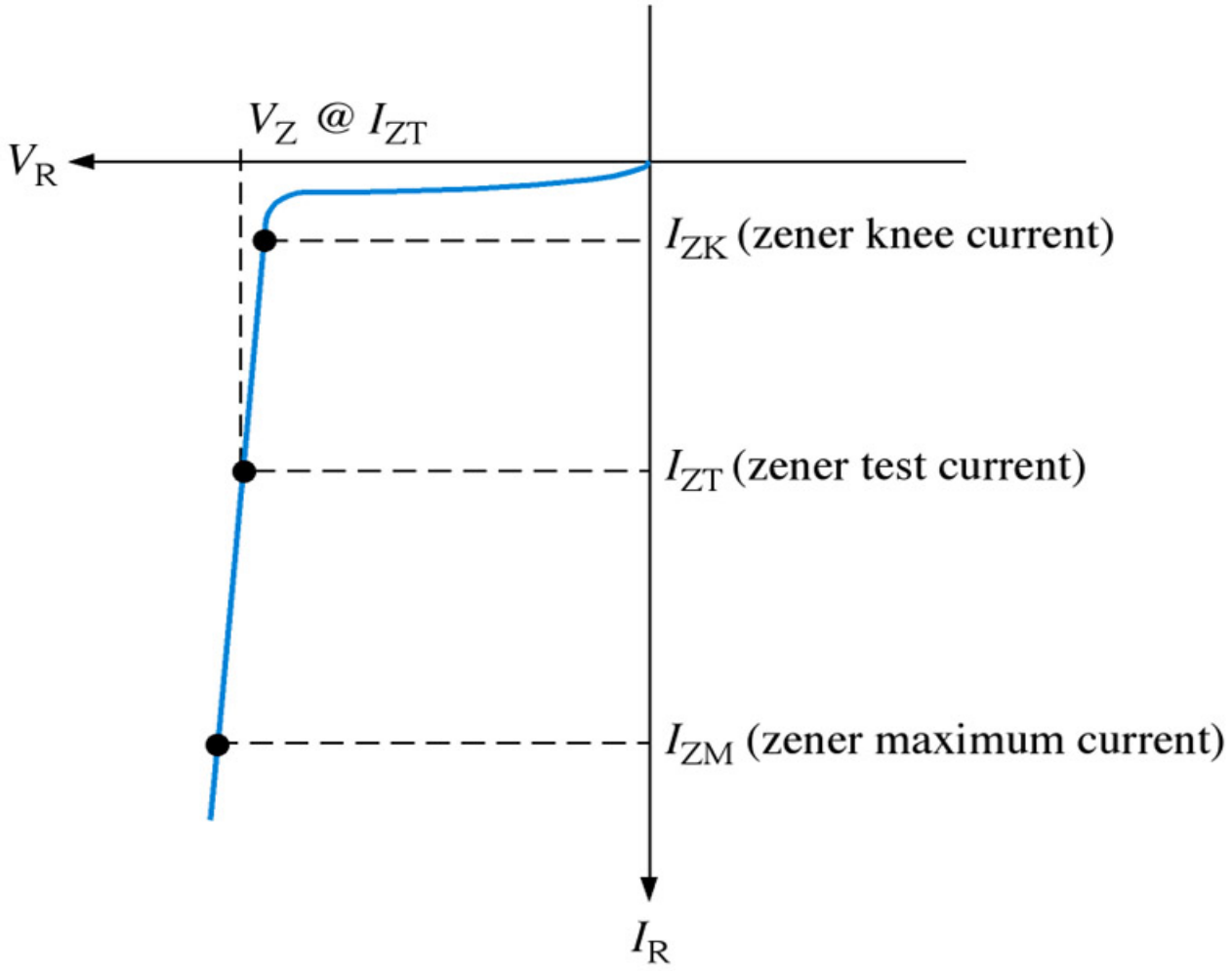


(a) The normal operating regions for a rectifier diode are shown as shaded areas.



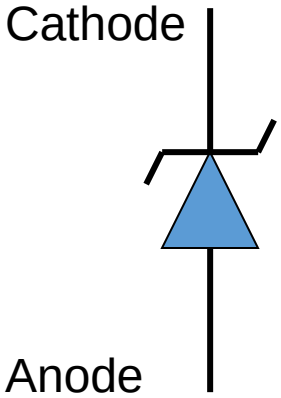
(b) The normal operating region for a zener diode is shaded.

Reverse characteristic of a zener diode. V_Z is usually specified at the zener test current, I_{ZT}, and is designated V_{ZT}.

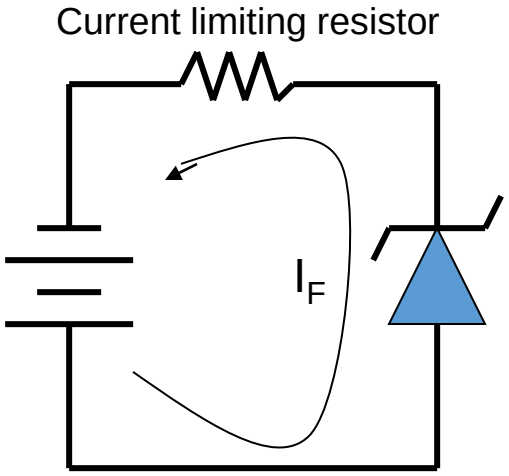


Circuit Symbol And Biasing Of A Zener Diode

- The circuit symbol of a zener diode is as shown in figure.
- The arrowhead in the symbol points towards the conventional direction of current through the diode, when it is forward bias.



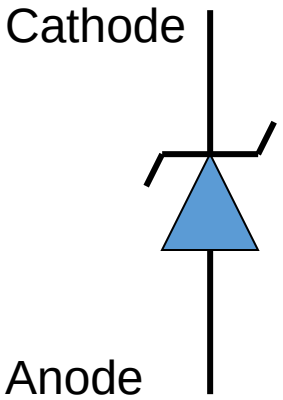
Circuit symbol



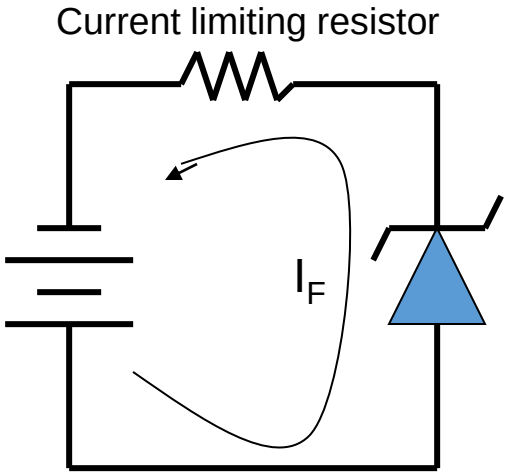
Forward biasing

Forward Biasing Of A Zener Diode

- When the anode of the zener diode is connected to the positive terminal of the DC source and the cathode is connected at the negative terminal, the zener diode is said to be forward bias.
- The forwards bias zener diode is behaves identical to a forward biased diode



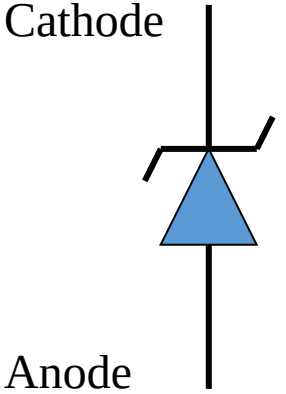
Circuit symbol



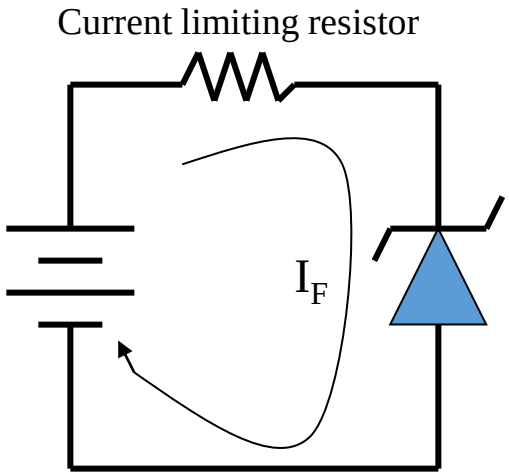
Forward biasing

Reverse Biasing Biasing Of A Zener Diode

- When a cathode is connected to the positive terminal and the anode is connected to the negative terminal of the DC source, the zener diode is said to be reverse bias.



Circuit symbol



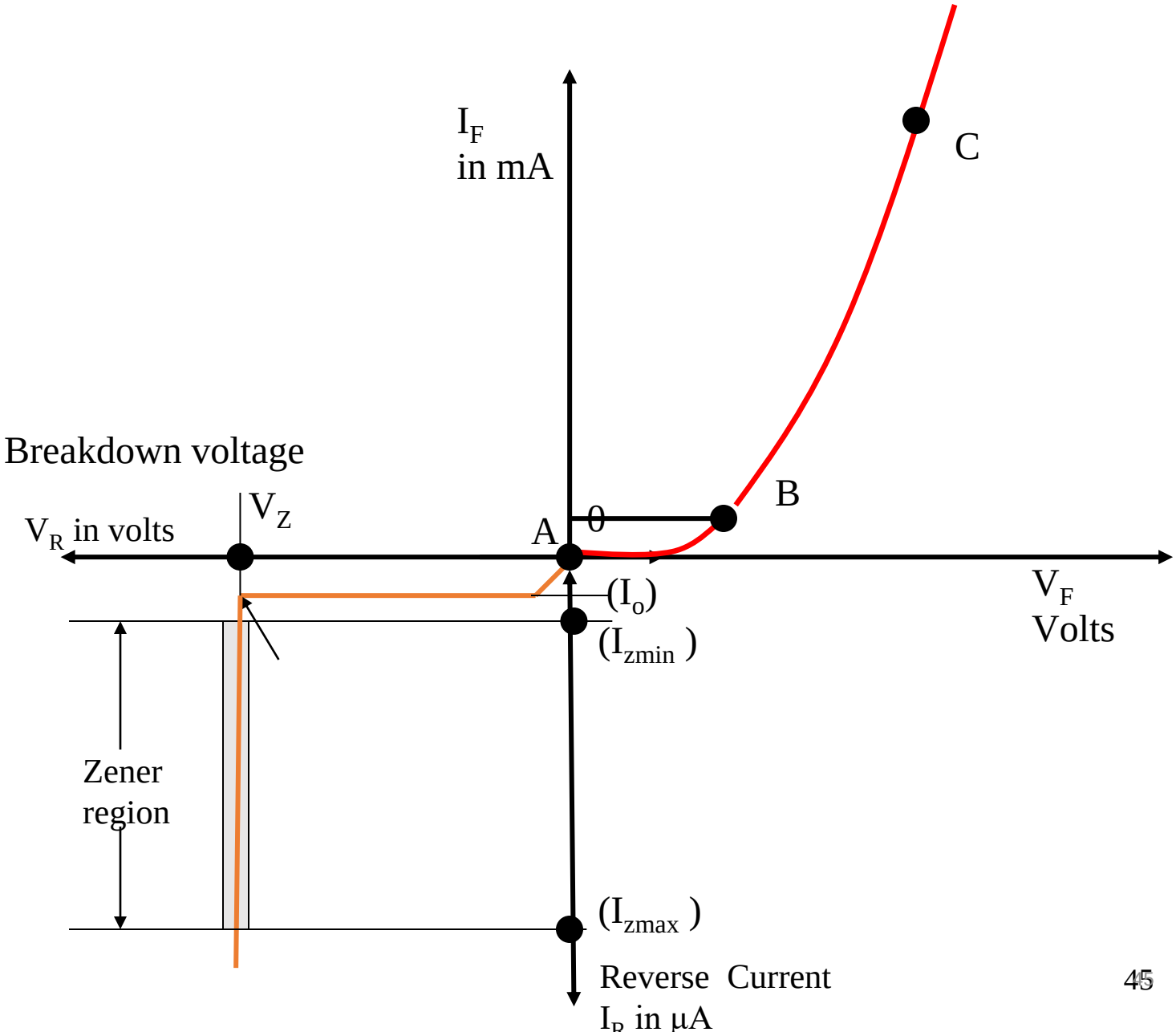
Reverse biasing 43

V-I characteristics of a Zener Diode

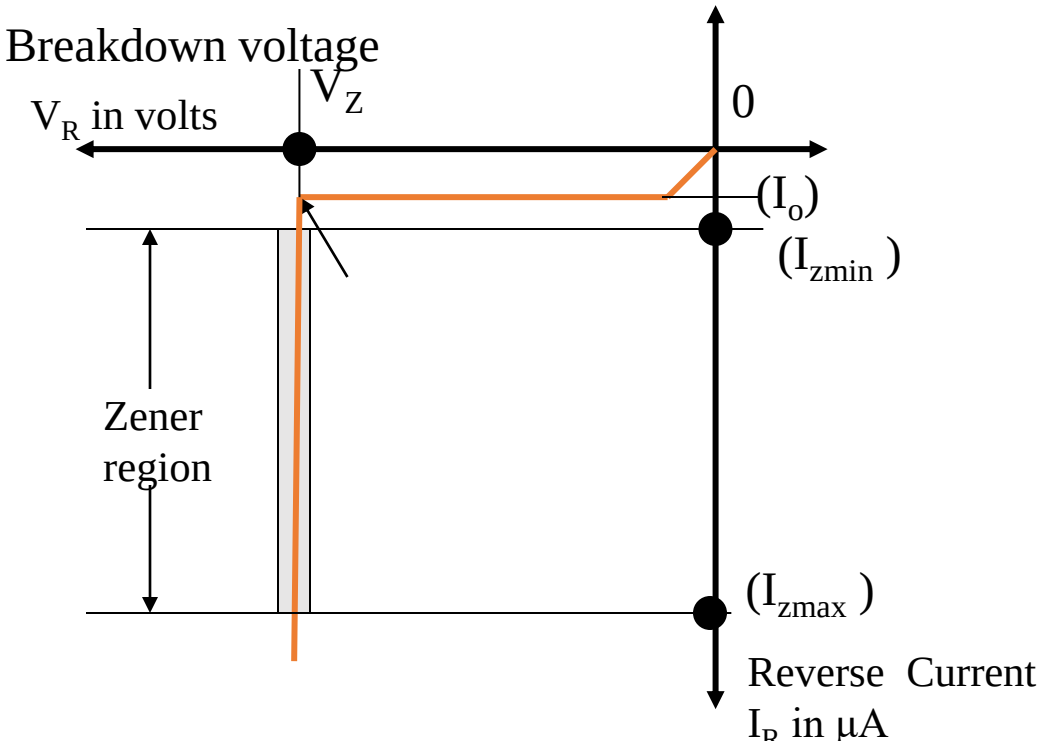
- The V-I characteristics of a zener diode can be divided into two parts.

- Forward characteristics
- Reverse characteristics.

Complete V-I Characteristics Of A Zener Diode



- Zener region :
- After reverse breakdown the zener diode operates in a region called zener region, as shown in figure.
- In this region the voltage across zener diode remains constant but current changes depending on the supply voltage.
- Zener diode operated in this region when it is being used as a voltage regulator



Breakdown Mechanism In The Zener Diode

- In the zener diode, there are two different breakdown mechanism, they are:
- Zener breakdown
- Avalanche breakdown

Breakdown Mechanism In The Zener Diode

- Zener breakdown :
- The zener breakdown is observed in the zener diodes having V_Z less than 5V or between 5 to 8 Volts.

- Avalanche breakdown :
- The avalanche breakdown is observed in the zener diodes having V_Z higher than 8 V.

Breakdown mechanism in the zener diode

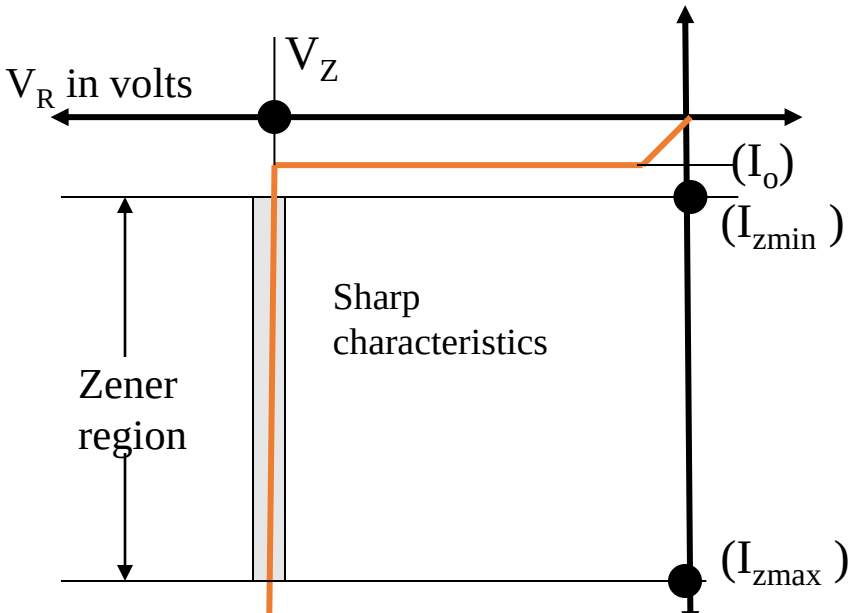
Zener breakdown :

- The zener breakdown is observed in the zener diodes having V_Z less than 5V or between 5 to 8 Volts.
- When a reverse voltage (5V or less) is applied to a zener diode, it cause a very intense electric field to appear across a narrow depletion region. The field intensity is typically of the order of $3X10^5$ V/cm.
- Such a intense electric field is strong enough to pull some of the valence electrons into the conduction band by breaking their covalent bonds. These electrons then become free electrons which are available for conduction.
- Large number of such free electrons will constitute a large reverse current through the zener diode and breakdown is said to have occurred due to the zener effect

Breakdown Mechanism In The Zener Diode

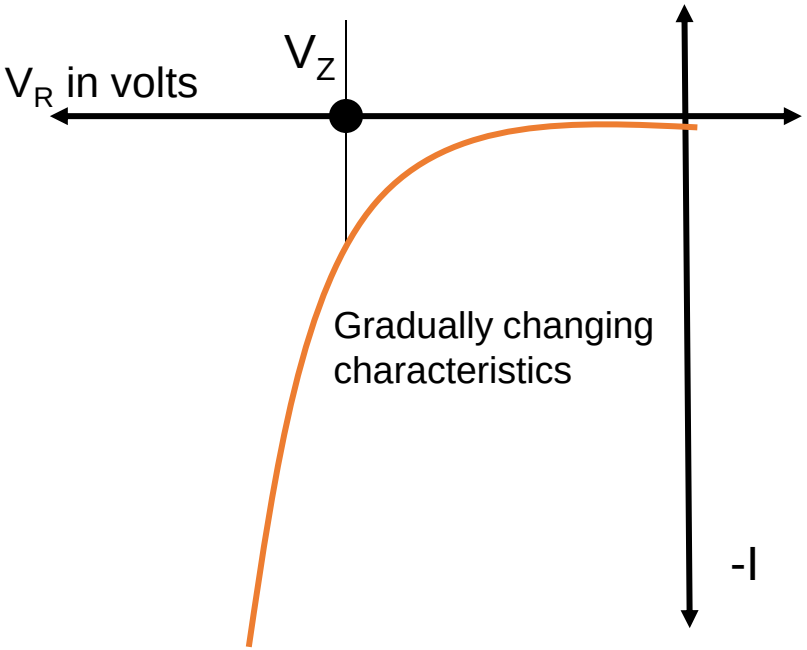
Zener breakdown :

- The current limiting resistor should be connected in series with the zener diode to protect it against the damage due to excessive heating.
- In zener breakdown the breakdown voltage depends on the temperature of p-n junction. The breakdown voltage decreases with increase in the junction temperature.
- The V-I characteristics in the reverse bias junction is shown in the figure:



Breakdown Mechanism In The Zener Diode

- Avalanche breakdown :
- The avalanche breakdown is observed in the zener diodes having V_Z higher than 8 V.

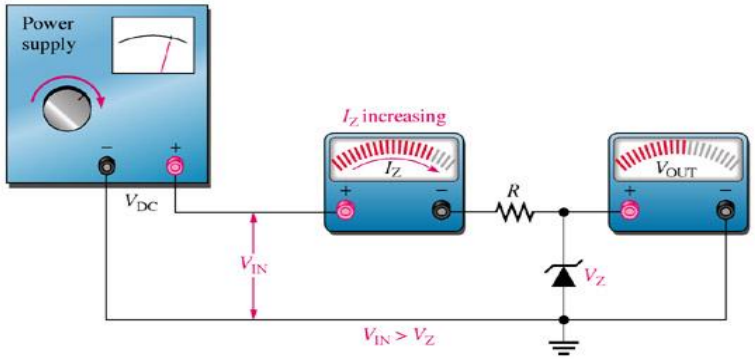


Comparison Of Zener And Avalanche Breakdown

Zener breakdown	Avalanche breakdown
This is observed in zener diodes having V_Z between 5 to 8 V	This is observed in zener diodes having V_Z greater than 8 V
The valence electrons are pulled into conduction band due to very intense electric field appearing across the narrow depletion region.	The valence electrons are pushed into conduction band due to the energy imparted by colliding accelerated minority carriers.
V-I characteristics with the zener breakdown is very sharp.	The V-I characteristics with the avalanche breakdown increases gradually. It is not sharp as that with the zener breakdown
The breakdown voltage decreases with increase in temperature	The breakdown voltage increases with increase in temperature.

Zener Diode Applications

- The Zener Diode is often used as a type of voltage regulator for providing stable reference voltage.



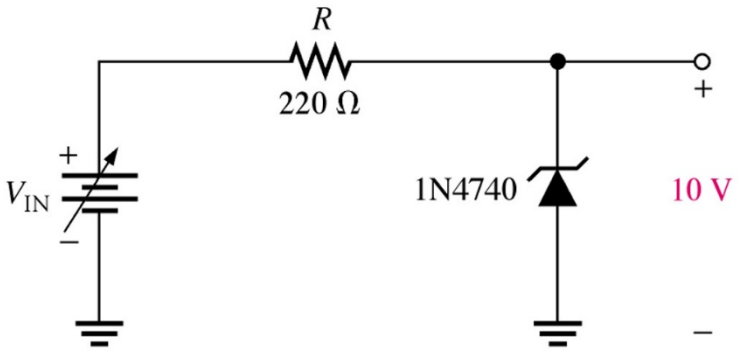
(a) As the input voltage increases, the output voltage remains constant ($I_{ZK} < I_Z < I_{ZM}$).

(b) As the input voltage decreases, the output voltage remains constant ($I_{ZK} < I_Z < I_{ZM}$).

Zener Regulation With A Varying Input Voltage

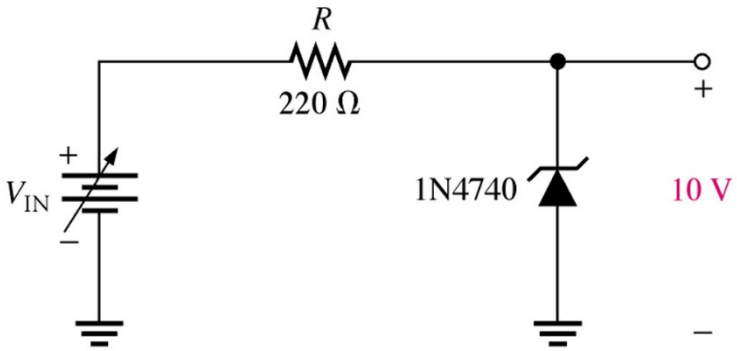
Zener Regulation With A Varying Input Voltage

- For example, suppose that the 1N4740 10V zener diode in figure can maintain regulation over a range of zener current values I_{ZK} =0.25mA to I_{ZM} =100mA.
- For minimum zener current, the voltage across the 220 Ω resistor is
- $V_R = I_{ZK}R = (0.25mA)(220\Omega) = 55mV$



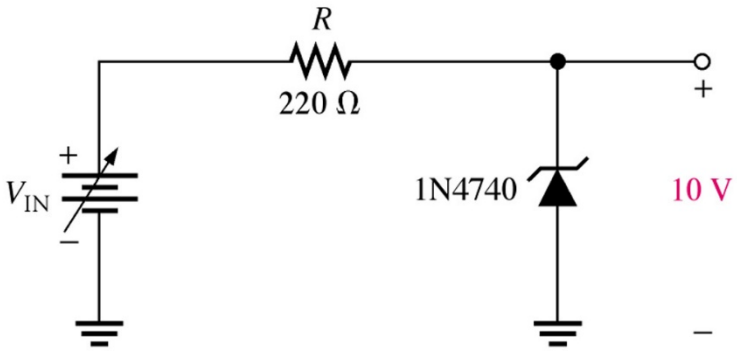
Zener Regulation With A Varying Input Voltage

- $V_R = I_{ZK}R = (0.25mA)(220\Omega) = 55mV$
- Since $V_R = V_{IN} - V_Z$
- $V_{IN(min)} = V_R + V_Z = 55mV + 10\ V = 10.055\ V$



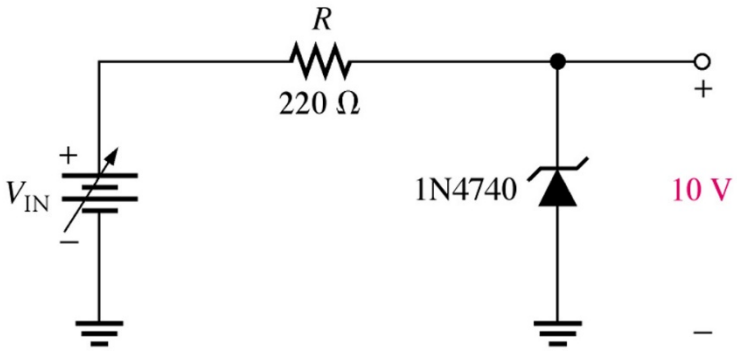
Zener Regulation With A Varying Input Voltage

- For the maximum zener current, the voltage across the 220Ω resistor is
- $V_R = I_{ZM}R = (100mA)(220\Omega) = 22V$
- Therefore
- $V_{IN(max)}=V_R + V_Z = 22\ V + 10\ V = 32\ V$



Zener Regulation With A Varying Input Voltage

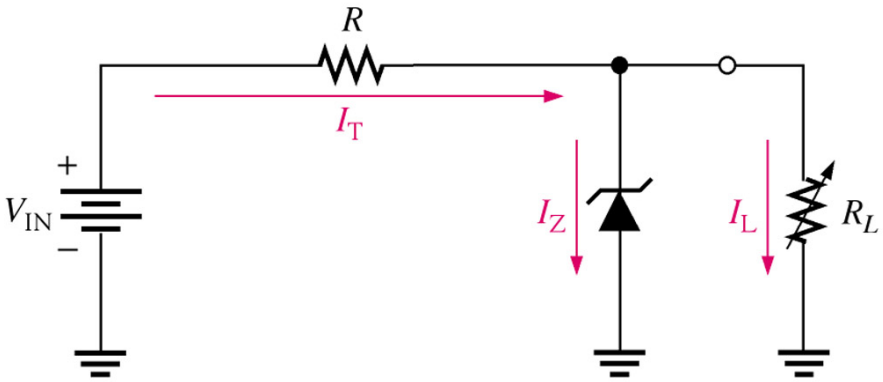
- This shows that this zener diode can regulate an input voltage from 10.55 V to 32V and maintain an approximate 10 V output



Zener Regulation With A Variable Load

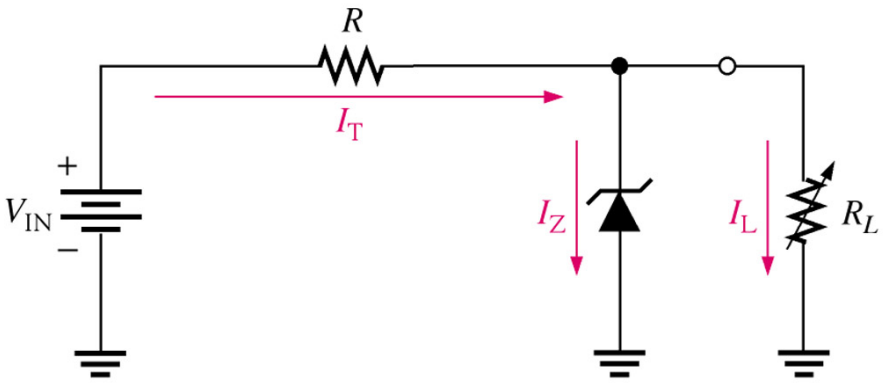
- Figure shows a zener voltage regulator with a variable load resistor across the terminals. The zener diode maintains a nearly constant voltage across R_L as long as the zener current is greater than I_{ZK} and less than I_{ZM}.

- <u>From No Load to Full Load :</u>
- When the output terminals of the zener regulator are open ($R_L = \infty$), the load current is zero and all of the current is through the zener.



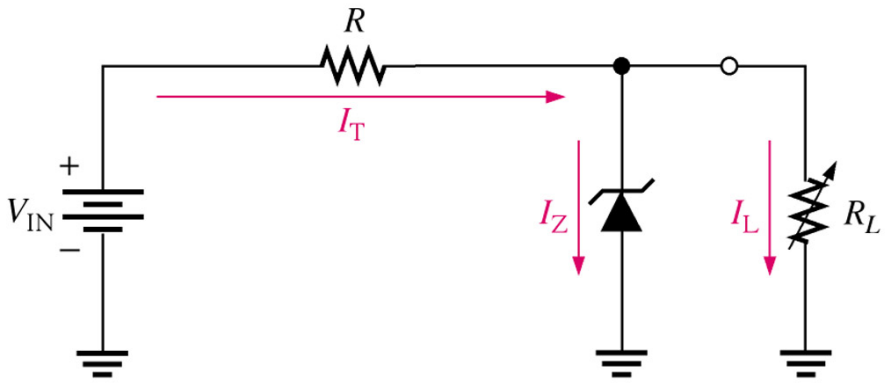
Zener Regulation With A Variable Load

- From No Load to Full Load :
- When a load resistor R_L is connected part of the total current is through the zener and part through R_L.
- As R_L is decreased, the load current, I_L increases and I_Z decreases.

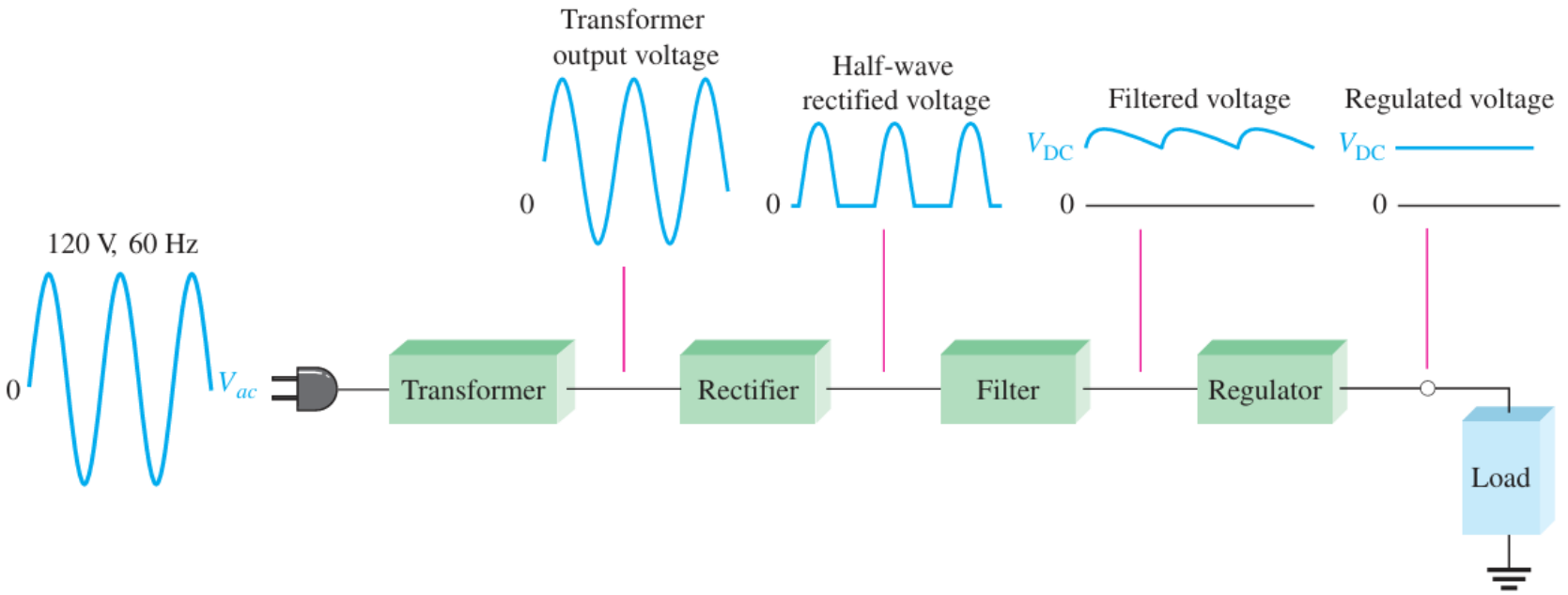


Zener Regulation With A Variable Load

- From No Load to Full Load :
- The zener diode continues to regulate the voltage until I_Z reaches its minimum value, I_{ZK}.
- At this point the load current is maximum.



Rectifiers and Filters



Complete power supply with transformer, rectifier, filter, and regulator

Rectifiers and Filters

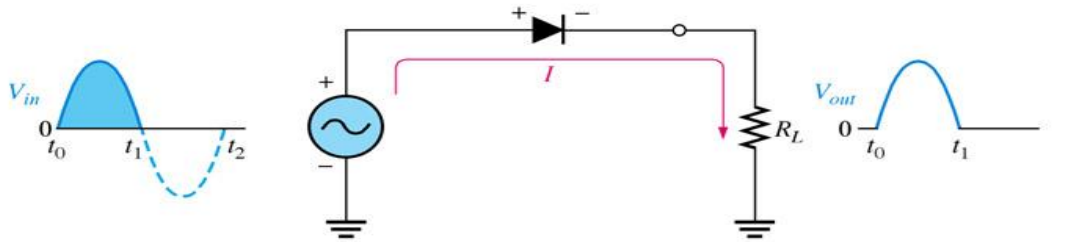
- The rectifiers and filters are the circuits, which are used in the initial stages of a d.c. power supply.
- D.C. power supply is essential for operation of many electronic devices and circuits.
- A rectifier is a circuit which uses one or more diodes to convert a.c. voltage into pulsating D.C. voltage.
- It may broadly categorized in the following two types.
  1. Half-wave Rectifier
  2. Full wave rectifier
     - A. With Center Tapped Transformer
     - B. Full Wave Bridge Rectifier

Analysis Of Rectifier

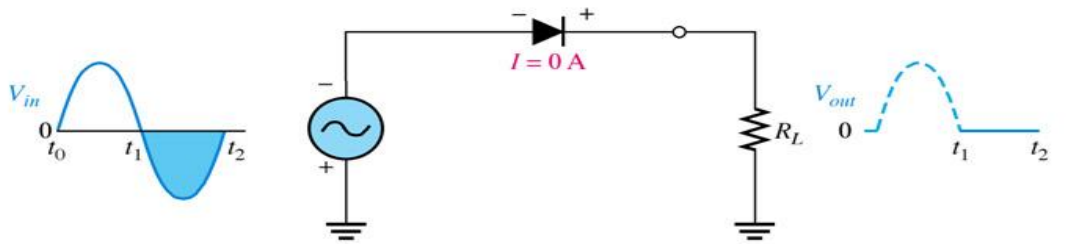
The important performance parameters of a rectifier are as follows :

- Average load current (I_{LDC})
- Average load voltage (V_{LDC})
- RMS load current (I_{Lrms})
- RMS load Voltage (V_{Lrms})
- Ripple factor
- Voltage regulation
- TUF and
- PIV

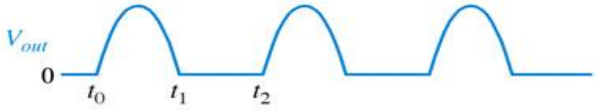
Half Wave Rectifier



(a) During the positive alternation of the 60 Hz input voltage, the output voltage looks like the positive half of the input voltage. The current path is through ground back to the source.



(b) During the negative alternation of the input voltage, the current is 0, so the output voltage is also 0.

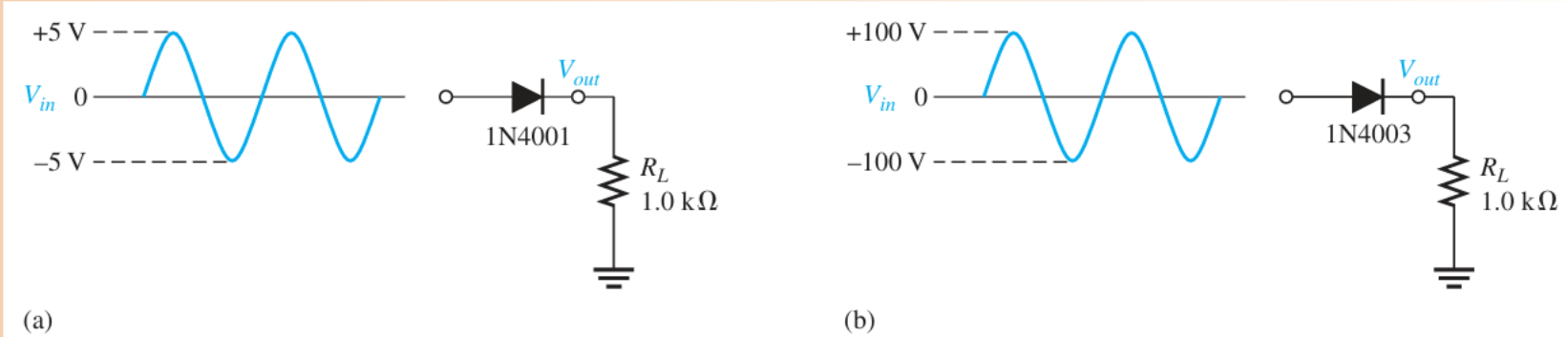


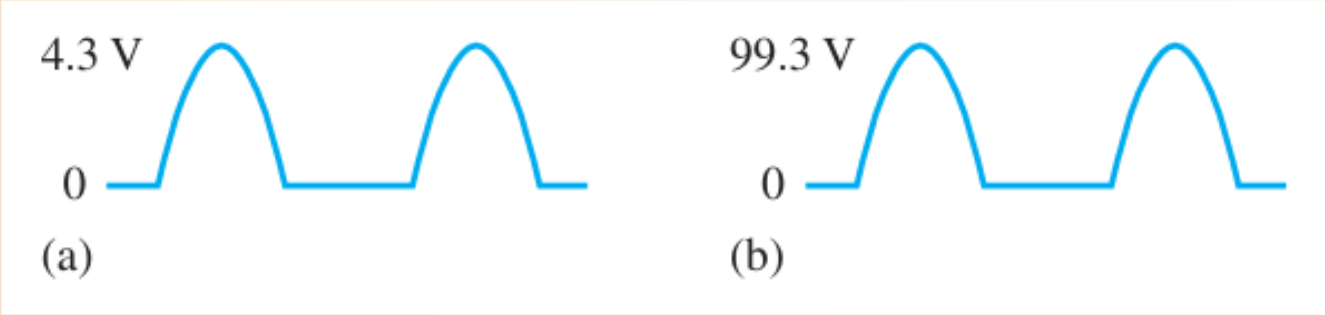
(c) 60 Hz half-wave output voltage for three input cycles

https://www.tinkercad.com/things/l2ppjnPesFe/editel

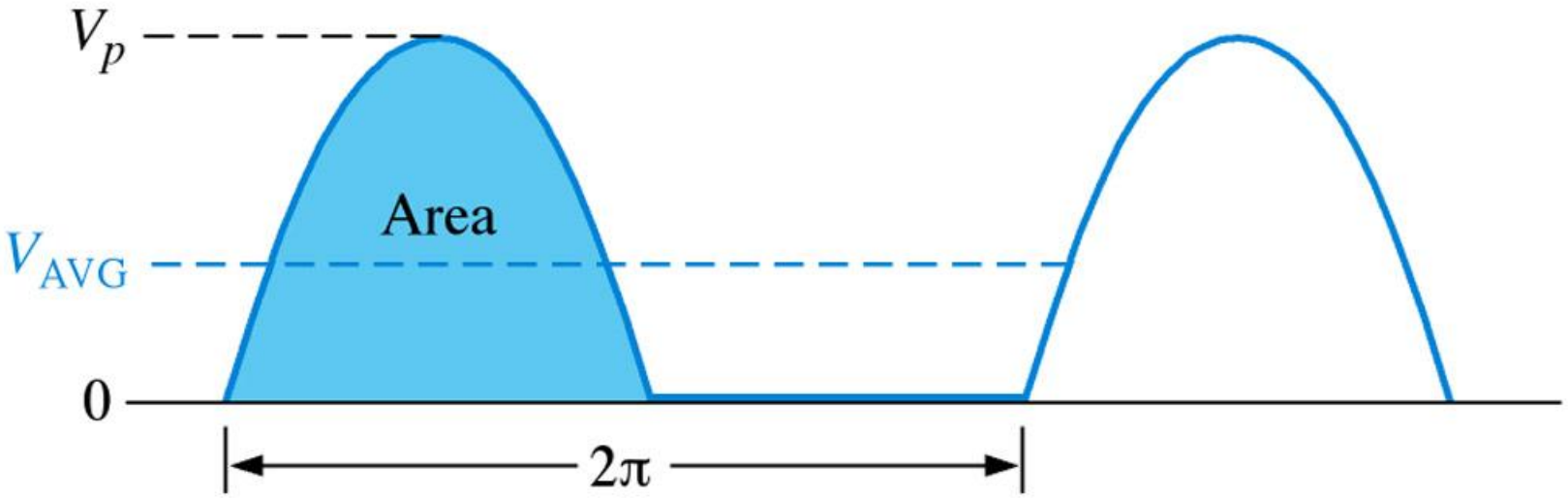
Half-wave rectifier operation. The diode is considered to be ideal.

EXAMPLE 2–3 Draw the output voltages of each rectifier for the indicated input voltages, as shown in Figure 2–24. The 1N4001 and 1N4003 are specific rectifier diodes.



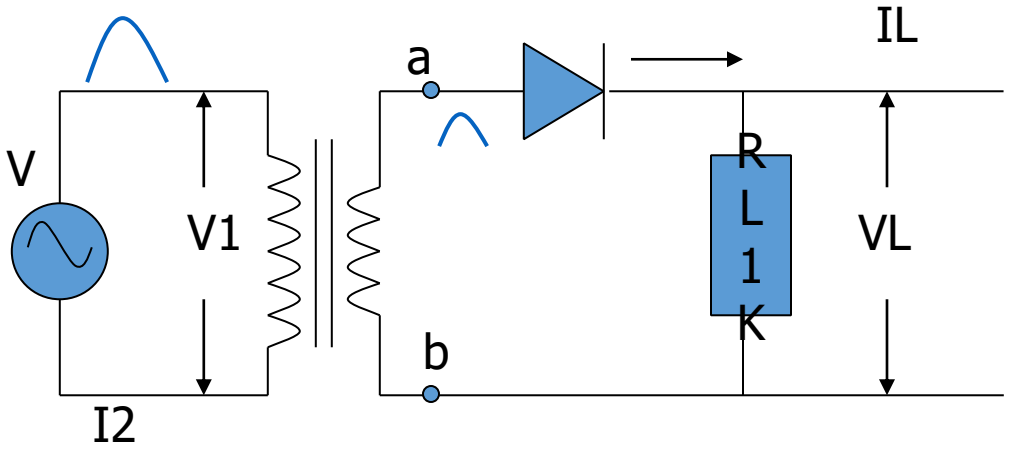


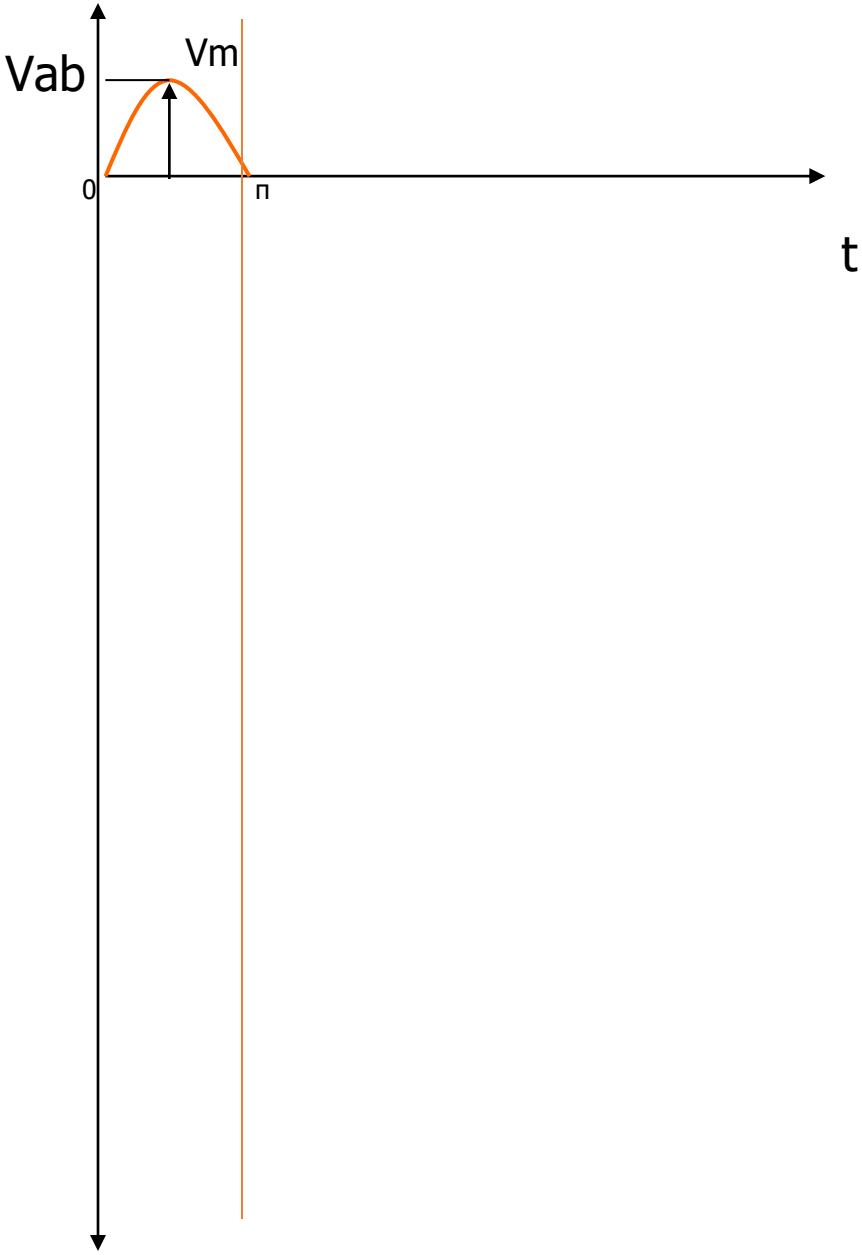
Half Wave Rectifier



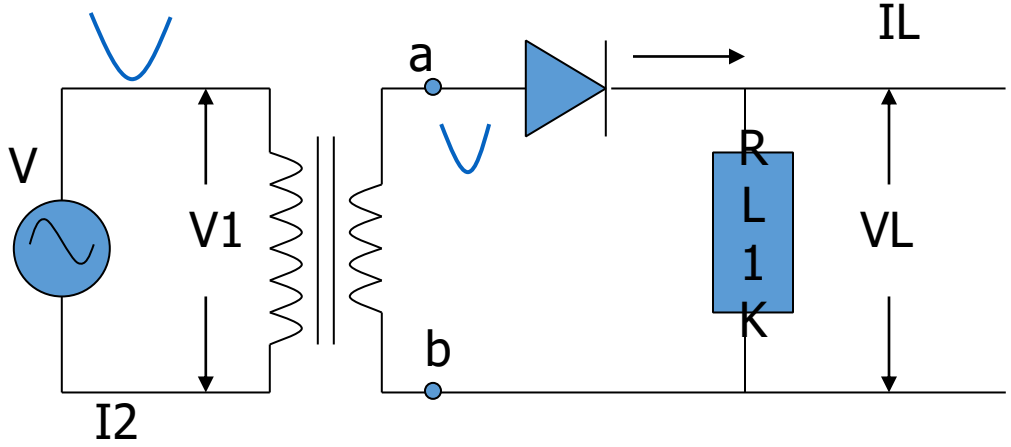
Average value of the half-wave rectified signal.

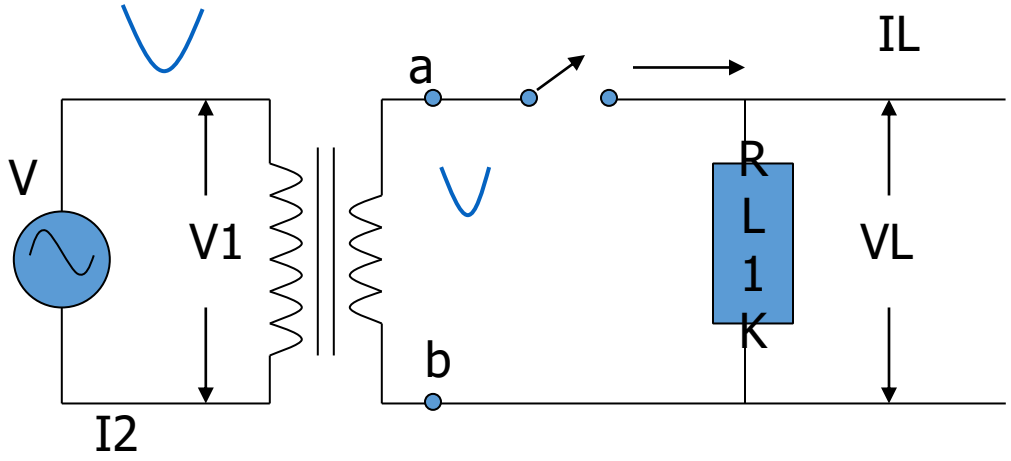
Half Wave Rectifier

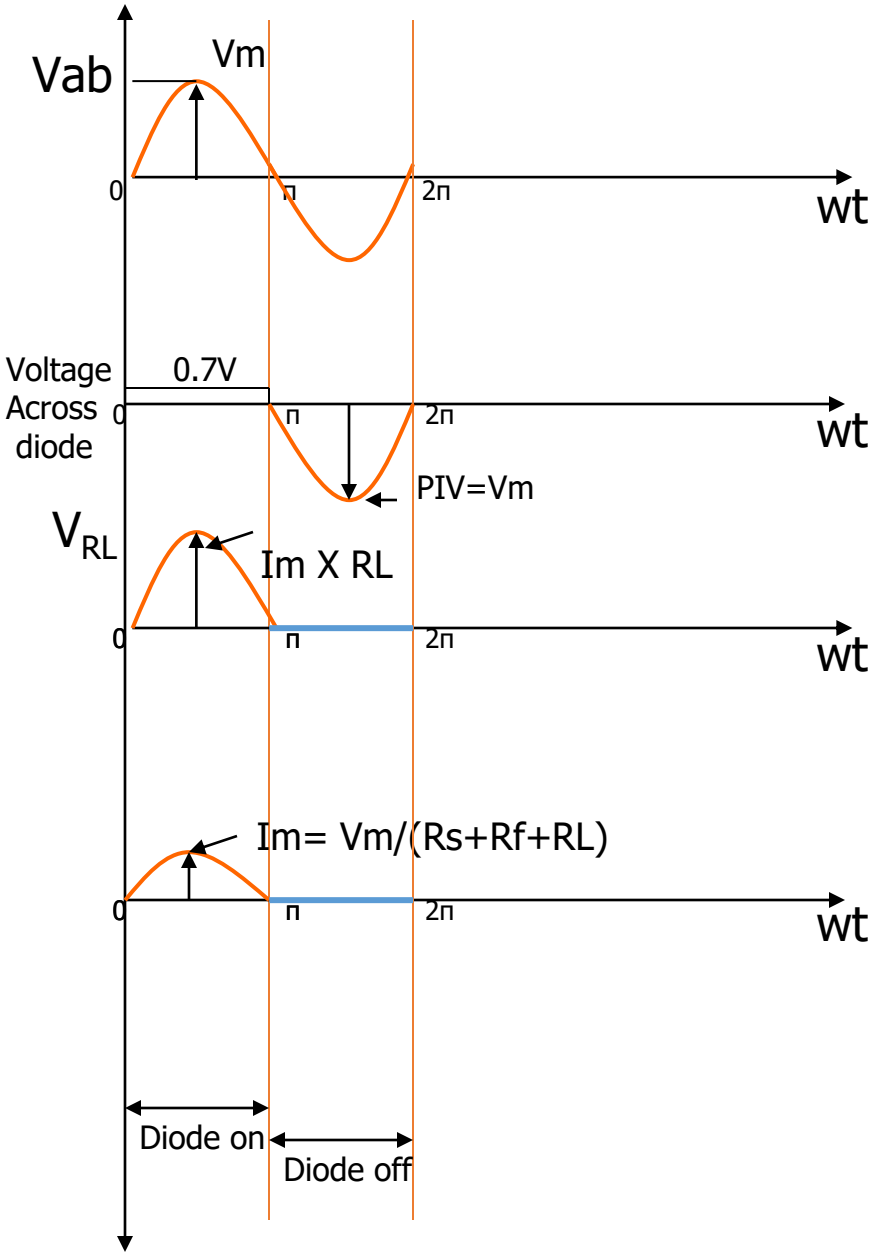




Half Wave Rectifier







Average values of output voltage and load current in a Half wave rectifier

$$V_{Ldc} = \frac{\int_0^{\pi} Vm \sin wt \, dwt}{2\pi}$$

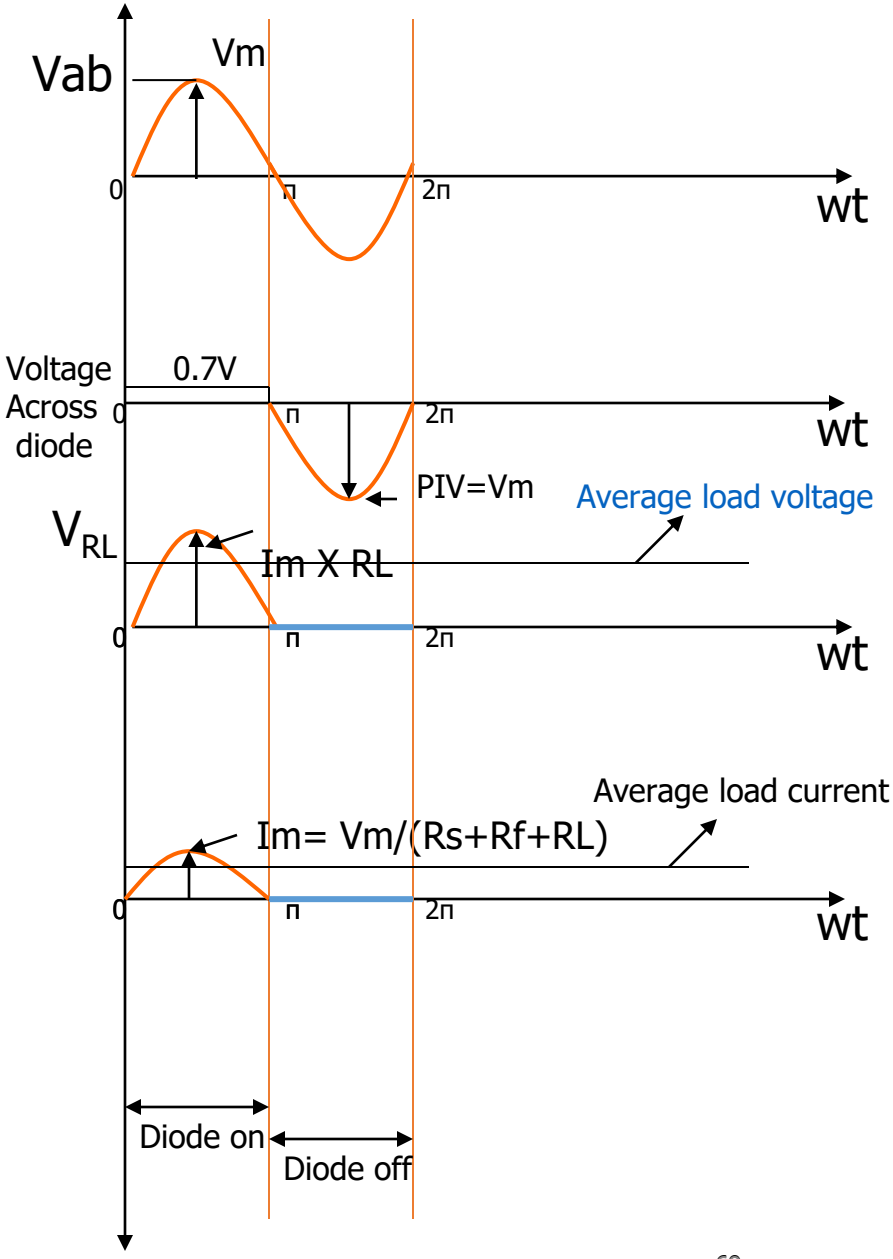
$$V_{LDC} = Vm/2\pi \; [-\cos wt]_0^{\pi}$$

$$= Vm/2\pi \; [+1-(-1)]$$

$$= 2Vm/2\pi$$

$$V_{LDC} = Vm/\pi$$

V_{LDC} = 0.318 Vm



Average values of output voltage and load current in a Half wave rectifier

$I_{LDC} = V_{LDC}/R_L$

$I_{LDC} = V_m/\pi R_L$

$I_{LDC} = I_m R_L/\pi R_L$

$I_{LDC} = I_m/\pi$

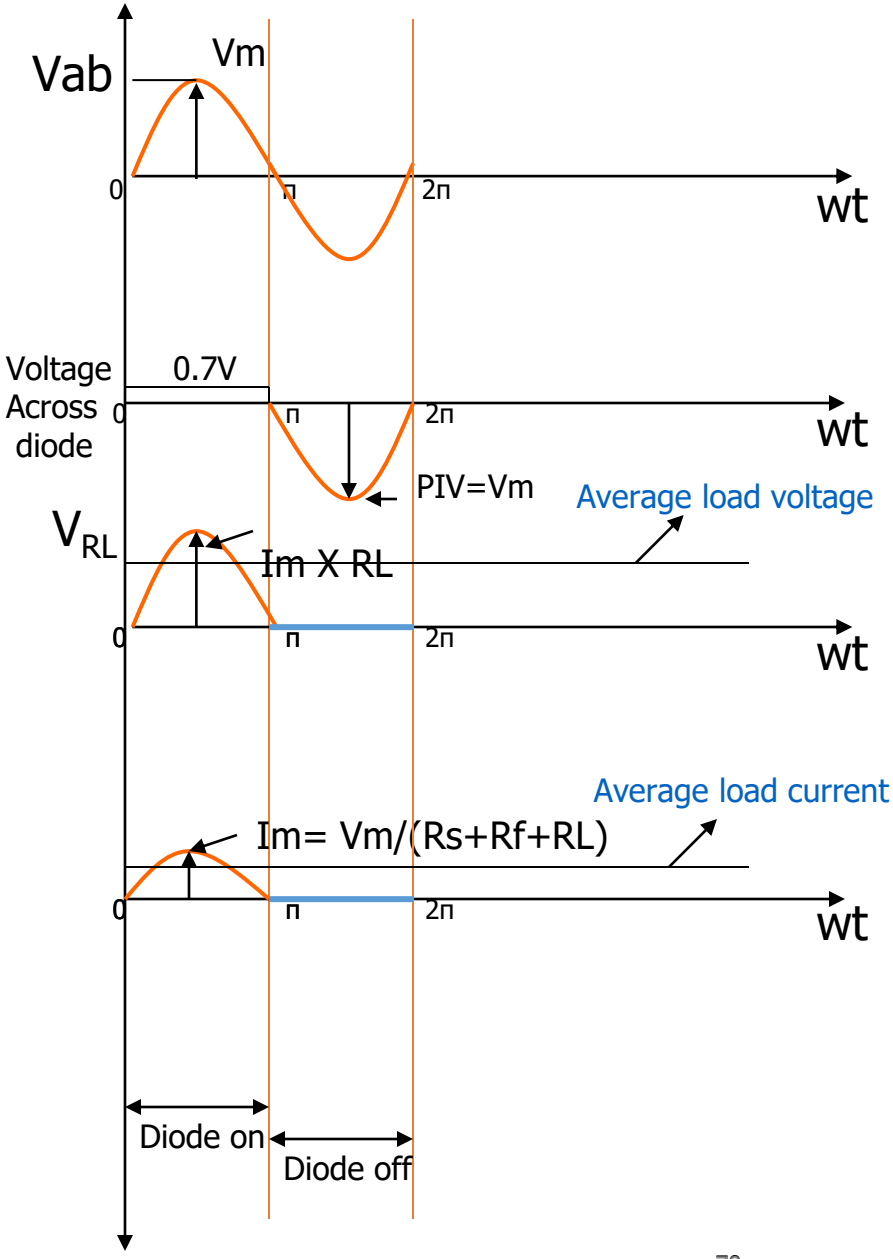
$V_{LDC} = 0.318\ Vm$

$V_m = Im/R_L$

$I_{LRMS} = Im/2$

$I_{LDC} = 0.318\ Im$

$V_{LRMS} = Vm/2$



Ripple Factor

- The rectifier output consist of AC as well as DC components. The ripple factor measures percentage of AC component in the rectifier output.
- The ripple factor indicates how close the rectified output is to the pure ideal dc voltage waveform.
- Small values of ripple factor indicates that the rectifier output waveform is close to being pure dc
- The ideal value of RF should be zero and practically it should be as small as possible.

- Ripple factor is denoted by “r”

$$\text{Ripple Factor} = \frac{\text{RMS value of the AC component of output}}{\text{DC or average value of the output}}$$

Ripple Factor

$$\text{Ripple Factor} = \frac{\text{RMS value of the AC component of output}}{\text{DC or average value of the output}}$$

$$r = \frac{[\, V^2_{LRMS} - V^2_{LDC} \,]^{1/2}}{V_{LDC}}$$

$$r = \frac{[\, (Vm/2)^2 - (Vm/\pi)^2 \,]^{1/2}}{Vm/\pi}$$

$$r = 1.21 \text{ or } 121\ \%$$

Equation indicates that the ripple content in the output voltage is 1.21 times the dc component. This is very high value of ripple factor which indicates that the output of HWR is no way close to the pure DC voltage

Voltage Regulation

- Ideally the average rectifier output voltage should remain constant. But practically it varies with the changes in the load current
- Voltage regulation is defined as:

- Voltage regulation = $\frac{V_{NL} - V_{FL}}{V_{FL}} \times 100\%$

Ideally the load regulation should be 0 % and practically it should be as low as possible.

DC Output Power

- The DC output power delivered to the load is given by:

- $P_{LDC} = I^2_{LDC}$ X RL

AC Output Power

- The AC output power delivered to the load is given by:

- $P_{LAC} = I^2_{LRMS}$ X (Rs +Rf + RL)

Rectifier Efficiency

- Rectifier efficiency is defined as,

- $$\eta = \frac{\text{DC output power}}{\text{AC input power}}$$

- Rectifier efficiency of a rectifier indicates the percentage of ac input power, actually converted into the average load power. Hence it should be as high as possible.

Peak Inverse Voltage (PIV) The maximum reverse voltage that a diode can withstand without destroying the junction is called peak inverse voltage of the diode.

PIV = Vm ... Half-wave rectifier

Transformer Utilization Factor (TUF)

- It may be defined as the ratio of DC power delivered to the load and the AC ratings of the transformer secondary.

- TUF = $\frac{\text{DC power delivered to the load}}{\text{AC rating of the transformer secondary}}$

$$= \frac{P_{DC}}{P_{AC}\ (\text{rated})}$$

For HWR the TUF = 0.287

For center tapped FWR the TUF = 0.693

And for bridge rectifier the TUF = 0.812

Disadvantages.

The main disadvantages of a half-wave rectifier are :

(i) The pulsating current in the load contains alternating component whose basic frequency is equal to the supply frequency. Therefore, an elaborate filtering is required to produce steady direct current.

(ii) (ii) The a.c. supply delivers power only half the time. Therefore, the output is low.

A diode having internal resistance rf = 20 Ω is used for half-wave rectification. If the applied voltage is v = 50 sin ωt and load resistance RL = 800Ω, find (i) Im, Idc, Ir.m.s. (ii) a.c. power input and d.c. power output (iii) d.c. output voltage and (iv) efficiency of rectification.

Solution. Maximum voltage, $V_m = 50$ V

(i)
$$I_m = \frac{V_m}{r_f + R_L} = \frac{50}{20 + 800} = 0.061 \text{ A} = \mathbf{61 \text{ mA}}$$
$$I_{dc} = I_m/\pi = 61/\pi = \mathbf{19.4 \text{ mA}}$$
$$I_{r.m.s.} = I_m/2 = 61/2 = \mathbf{30.5 \text{ mA}}$$

(ii)
$$P_{ac} = I^2_{r.m.s.}\,(r_f + R_L)$$
$$= (30.5 \times 10^{-3})^2 \times (20 + 800) = \mathbf{0.763 \text{ W}}$$

$$P_{dc} = I^2_{dc} \times R_L = (19.4 \times 10^{-3})^2 \times 800 = \mathbf{0.301 \text{ W}}$$

(iii) D.C. output voltage $= I_{dc} \times R_L = 19.4 \text{ mA} \times 800\ \Omega = \mathbf{15.52 \text{ V}}$

Alternatively, D.C. output voltage $= V_m/\pi = 50/\pi = 15.52$ V

(iv)
$$\text{Efficiency} = \frac{0.301}{0.763} \times 100 = \mathbf{39.5\%}$$

Example 1

- An a.c. supply of 200 v is applied to a half wave rectifier circuit through a transformer of turns ratio 10:1, find
- d.c. output voltage
- PIV of a diode, assume that diode is an ideal one.

Example

- An a.c. supply of 200 v is applied to a half wave rectifier circuit through a transformer of turns ratio 10:1, find
- d.c. output voltage
- PIV of a diode, assume that diode is an ideal one.

- Ans:
- d.c. output voltage

we know that the secondary voltage,

$V2 = V1 * (N2/N1) = 23V$

The maximum value of secondary voltage

$Vm = \sqrt{2} * V2 = 32.5\ V$

d.c. output voltage $Vdc = 0.318\ Vm = 10.3\ V$

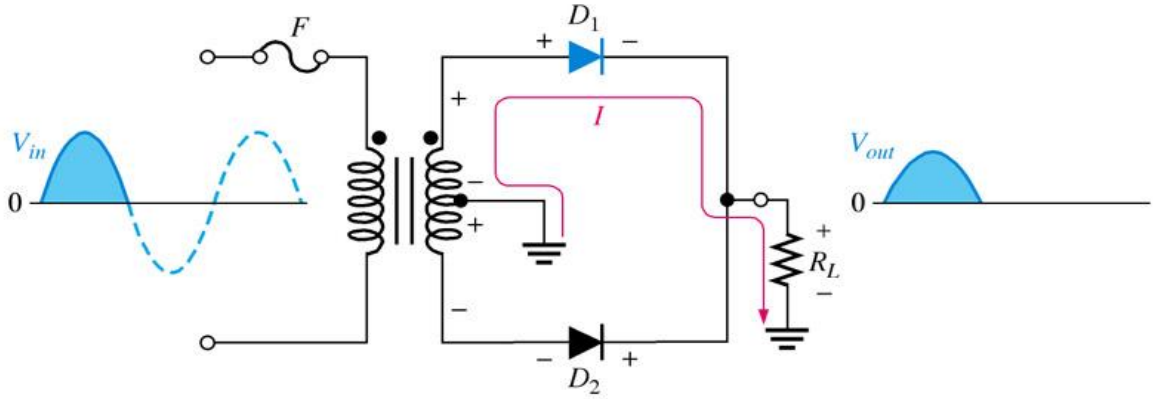
- PIV of a diode

We also know the peakk inverse voltage

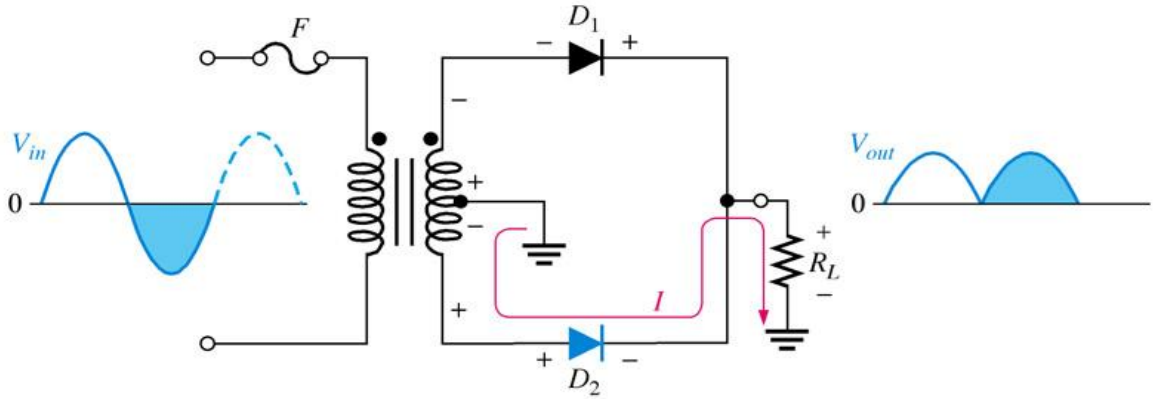
$PIV = Vm = 32.5V$

Full Wave Rectifier With Centre Tapped Transformer

Basic operation of a center-tapped full-wave rectifier. Note that the current through the load resistor is in the same direction during the entire input cycle, so the output voltage always has the same polarity.

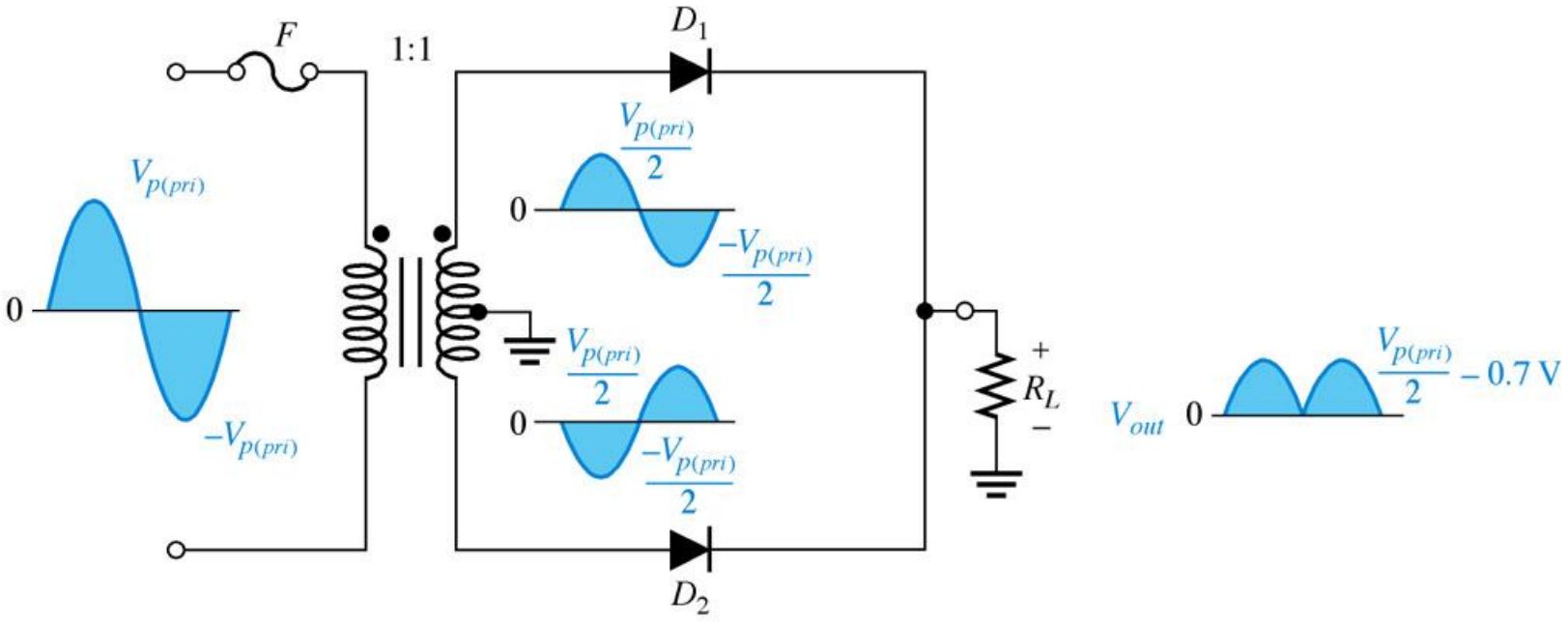


(a) During positive half-cycles, D_1 is forward-biased and D_2 is reverse-biased.

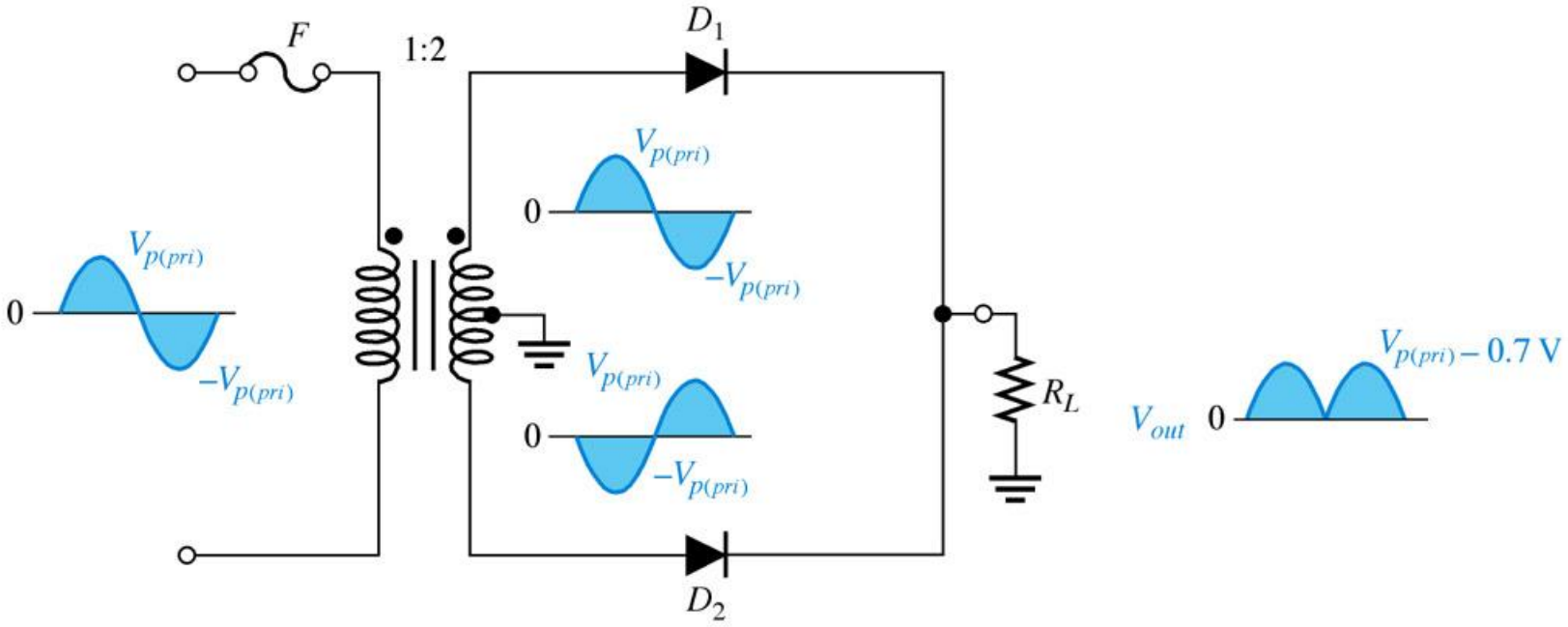


(b) During negative half-cycles, D_2 is forward-biased and D_1 is reverse-biased.

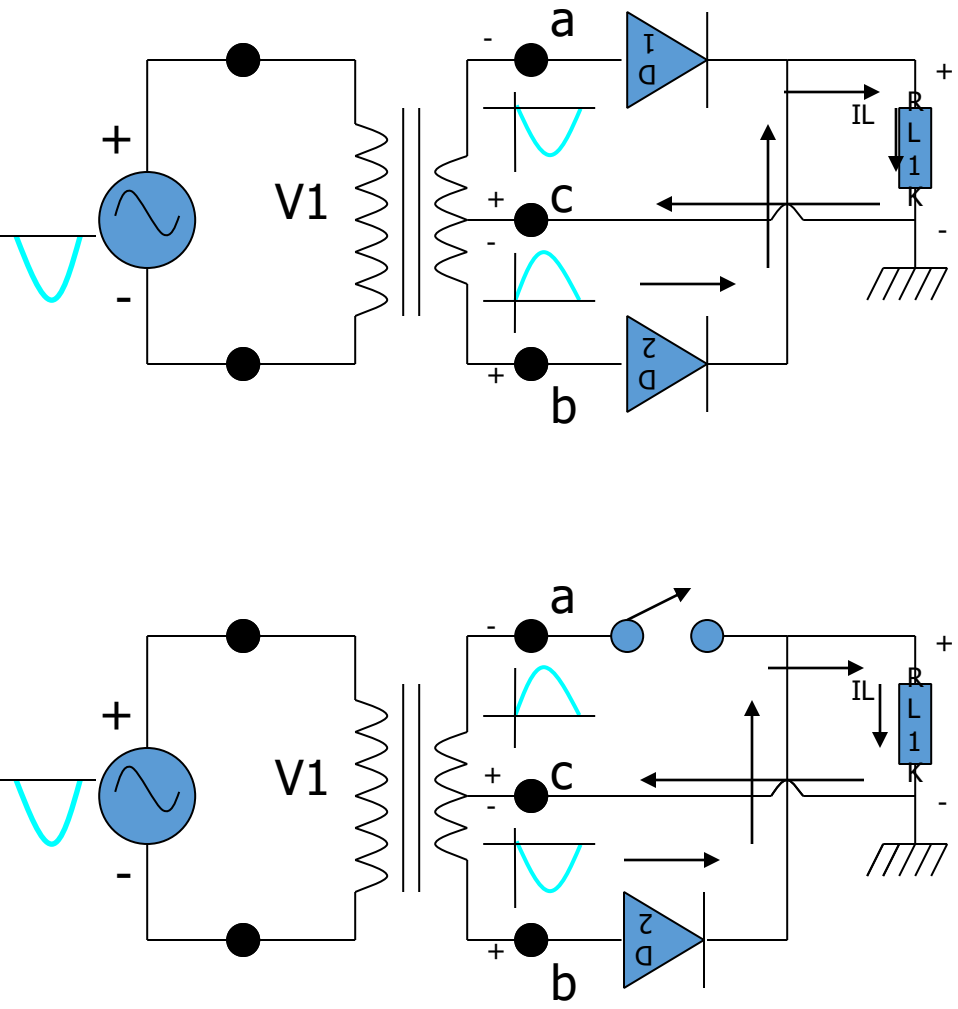
Center-tapped full-wave rectifier with a transformer turns ratio of 1. $V_{p(pri)}$ is the peak value of the primary voltage.

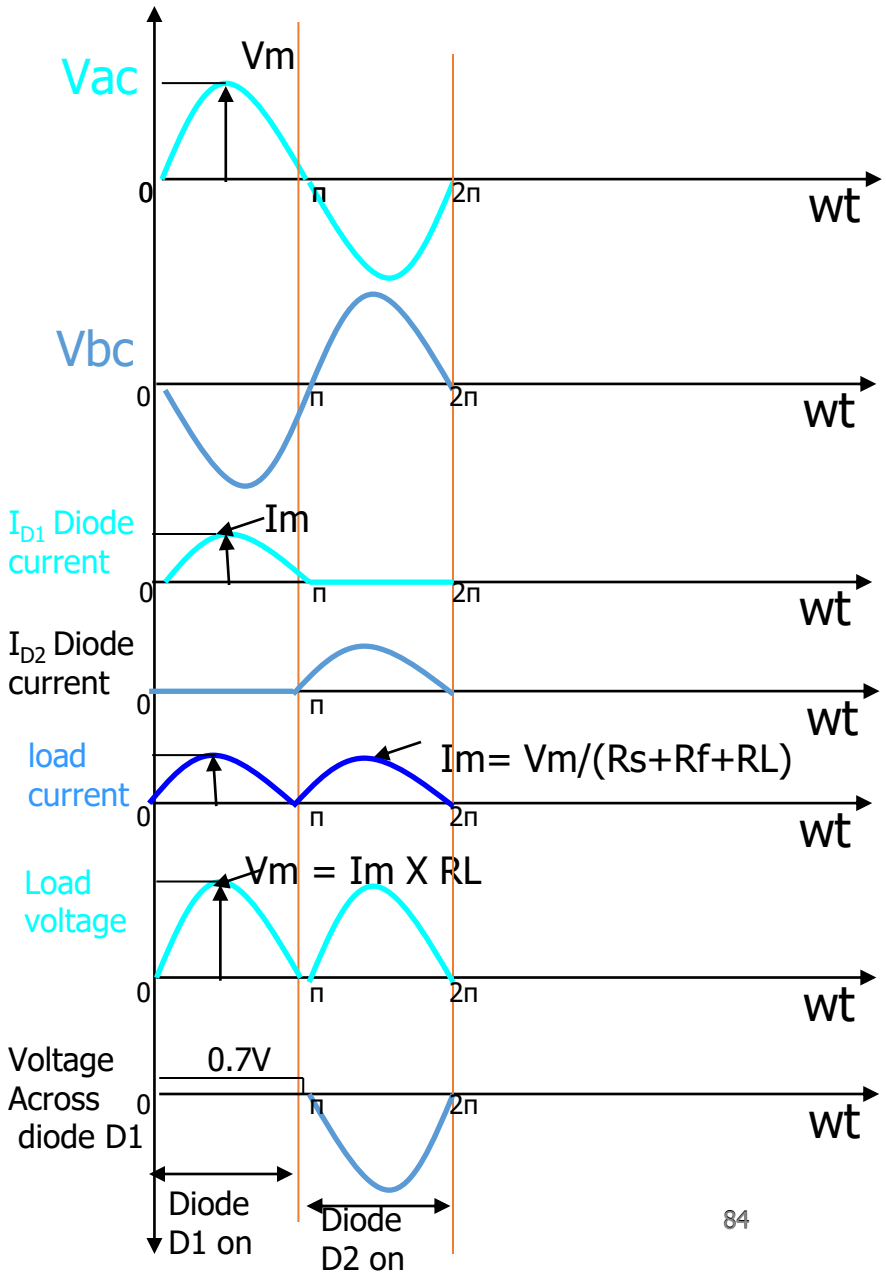


Center-Tapped Full-Wave Rectifier With A Transformer Turns Ratio Of 2.



Full Wave Rectifier With Centre Tapped Transformer





Average values of output voltage and load current in a full wave rectifier

instantaneous value of the sinusoidal a.c. input is given by the relation

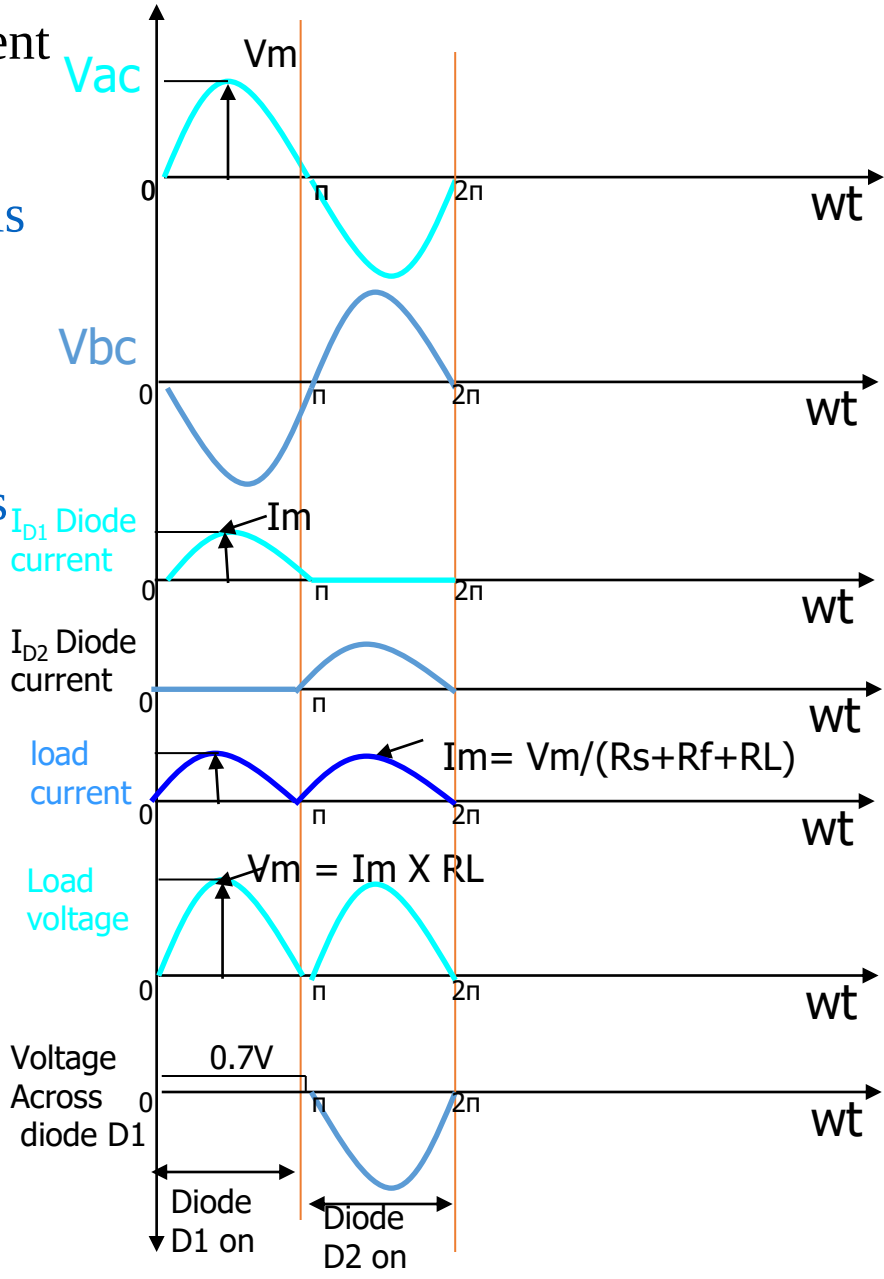
$v = Vm \sin wt$----------(1)

the average or d.c. value of the output voltage is given by the relation

V_{Ldc}= area under the curve over the full cycle

base

$$V_{Ldc} = \frac{\int_0^{\pi} v \, dwt}{\pi}$$



Average values of output voltage and load current in a full wave rectifier

$$V_{Ldc} = \frac{\int_0^{\pi} Vm \sin wt \, dwt}{\pi}$$

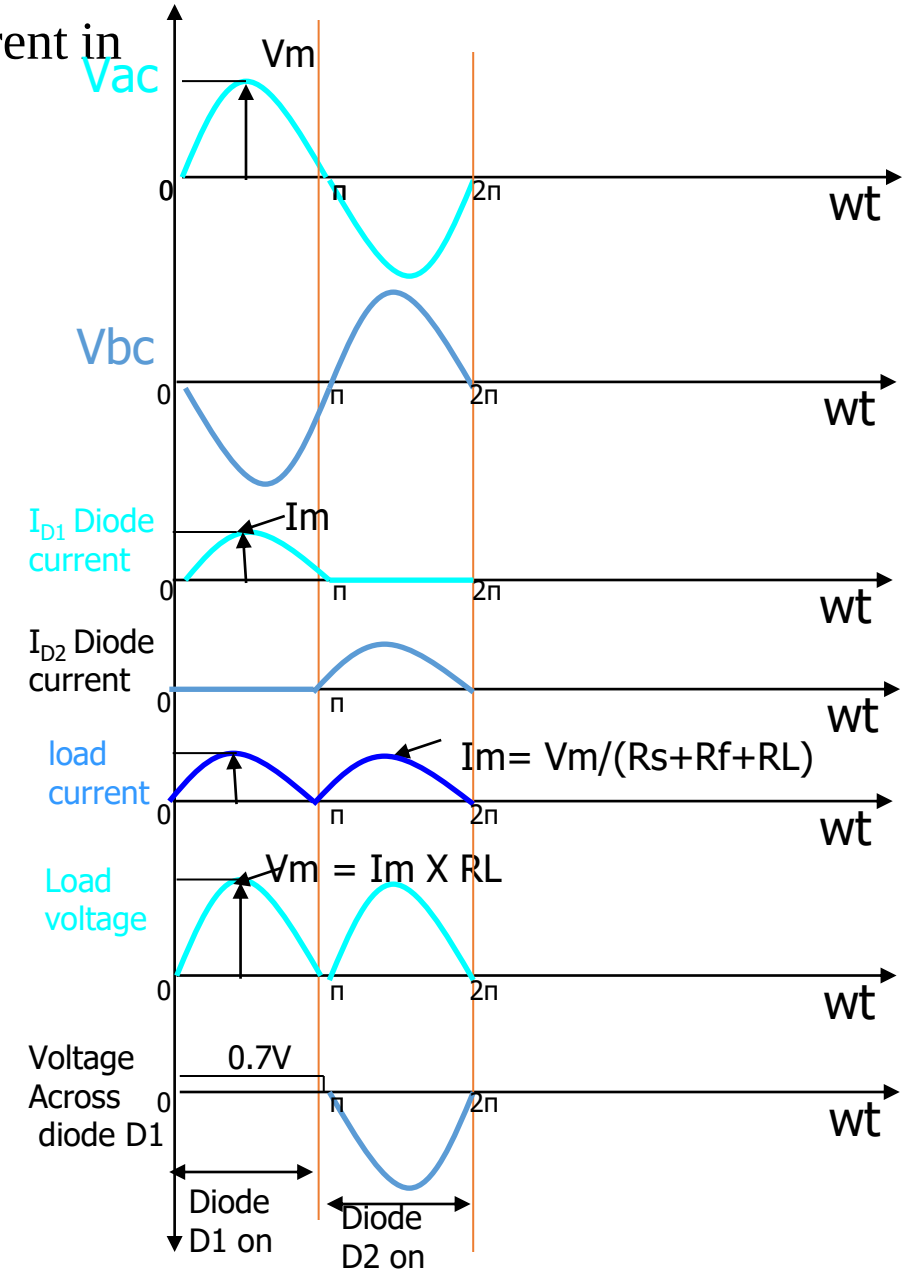
$$V_{LDC} = Vm/\pi \; [-\cos wt]_0^{\pi}$$

$$= Vm/\pi \; [+1-(-1)]$$

$$= 2Vm/\pi$$

$$V_{LDC} = 2Vm/\pi$$

$V_{LDC} = 2Vm/\pi = 0.636\ Vm$



Average values of output voltage and load current in a full wave rectifier

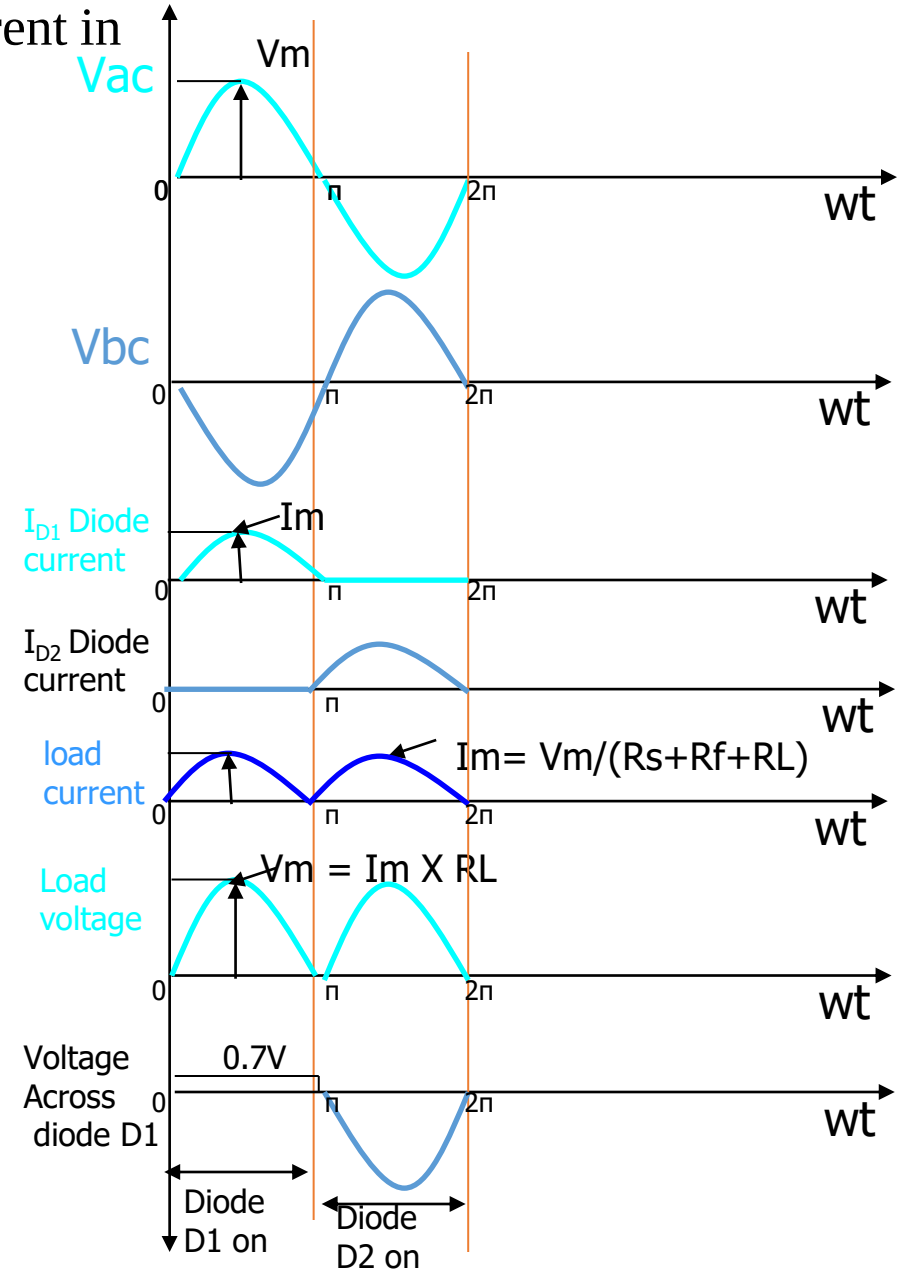
$I_{LDC} = V_{LDC}/RL$

$= (2Vm/\pi)RL$

$= (2Im * RL/\pi) * RL$

$I_{LDC} = 2Im/\pi$

$I_{LDC} = 2Im/\pi = 0.636\ Im$

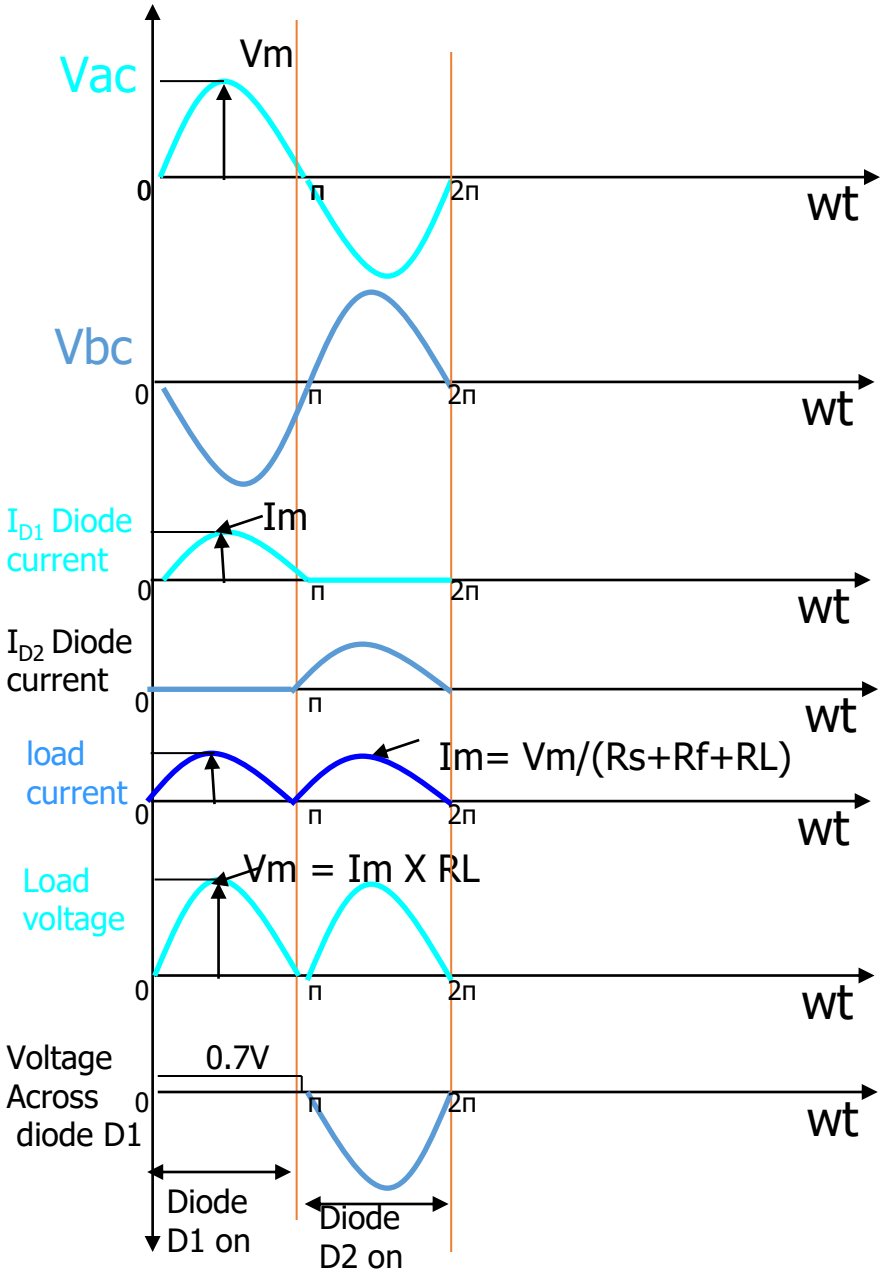


RMS or AC load current

$$I_{LRMS} = Im/\sqrt{2}$$

$$I_{LRMS} = Im/\sqrt{2}$$

$$V_{LRMS} = Vm/\sqrt{2}$$



Ripple factor

$$\text{ripple factor} = \frac{\text{RMS value of the AC component of output}}{\text{DC or average value of the output}}$$

$$r = \frac{[\, V^2_{LRMS} - V^2_{LDC} \,]^{1/2}}{V_{LDC}}$$

$$r = \frac{[\, (Vm/\sqrt{2})^2 - (2Vm/\pi)^2 \,]^{1/2}}{2\, Vm/\pi}$$

$r = 0.48$ or 48 %

DC Output Power

- The DC output power delivered to the load is given by:

- $P_{LDC} = I^2_{LDC}$ X RL

AC Input Power

- The AC input power:

- $P_{AC} = I^2_{LRMS}$ X RL

DC output power

- The DC output power delivered to the load is given by:

- $P_{LDC} = I^2_{LDC}$ X R_L

substituting $I_{LDC} = 2I_m / \pi$ we get

$$P_{LDC} = (4I^2_m / \pi^2) \text{ X } R_L \qquad \text{since } I_M = V_M / (R_S + R_F + R_L)$$

$$P_{LDC} = [4V^2_m / \pi^2 (R_S + R_F + R_L)] \text{ X } R_L$$

$$\boxed{P_{LDC} = [4V^2_m / \pi^2 (R_S + R_F + R_L)^2] \text{ X } R_L}$$

AC Input Power

- The AC input power:
- $P_{AC} = I^2_{S\,RMS}$ X $(R_S + R_F + R_L)$

but $I_{SRMS} = I_M /\sqrt{2}$

- $P_{AC} = (I_M /\sqrt{2})^2$ X $(R_S + R_F + R_L)$
- $P_{AC} = [I_M{}^2$ X $(R_S + R_F + R_L)] / 2$

Since $I_M = V_M /(R_S + R_F + R_L)$

- $P_{AC} = [V_M{}^2$ X $(R_S + R_F + R_L)] / 2 (R_S + R_F + R_L)^2$
- $P_{AC} = V_M{}^2 / 2 (R_S + R_F + R_L)$

$$\mathbf{P_{AC} = V_M{}^2 / 2 (R_S + R_F + R_L)}$$

Rectifier Efficiency

- Rectifier efficiency is defined as,

- $$\eta = \frac{\text{DC output power}}{\text{AC input power}}$$

- $$\eta = P_{LDC}/P_{AC}$$

$$\text{Since } P_{LDC} = I^2_{LDC} \quad X \quad R_L$$
$$P_{AC} = I^2_{S\,RMS} \quad X \quad (R_S + R_F + R_L)$$

- $$\eta = [I^2_{LDC} \quad X \quad R_L] / [I^2_{S\,RMS} \quad (R_S + R_F + R_L)]$$

- $$\text{since} \quad I_{LDC} = 2I_m/\pi \quad \text{and} \quad I_{SRMS} = I_M/\sqrt{2}$$
- $\eta = 0.812$ or 81.2 %

- This is the maximum theoretical value of rectifier efficiency.
- Rectifier efficiency of FWR is almost twice the rectifier efficiency of HWR.

Transformer Utilization Factor (TUF)

- It may be defined as the ratio of DC power delivered to the load and the AC ratings of the transformer secondary.

- TUF of secondary winding = $\dfrac{\text{DC power delivered to the load}}{\text{AC rating of the transformer secondary}}$

$$= \frac{P_{DC}}{P_{AC}\,(\text{rated})}$$

$$= \frac{[\, I_{LDC} \text{ X } V_{LDC} \,]}{[\, I_{SRMS} \text{ X } V_{SRMS} \,]}$$

Transformer Utilization Factor (TUF)

- It may be defined as the ratio of DC power delivered to the load and the AC ratings of the transformer secondary.

- TUF of secondary winding = $\dfrac{\text{DC power delivered to the load}}{\text{AC rating of the transformer secondary}}$

$$= \frac{P_{DC}}{P_{AC}\ (\text{rated})}$$

$$= \frac{[\ I_{LDC}\ X\ V_{LDC}\ ]}{[\ I_{SRMS}\ X\ V_{SRMS}\ ]}$$

secondary TUF = 0.812 or 81.2 %

Transformer Utilization Factor (TUF)

It may be defined as the ratio of DC power delivered to the load and the AC ratings of the transformer secondary.

A full wave rectifier consists of two half wave rectifiers, working independently and feeding a common load.

Therefore Primary TUF = 2 X TUF of HWR = 2 X 0.287

= 0.574

Transformer Utilization Factor (TUF)

- It may be defined as the ratio of DC power delivered to the load and the AC ratings of the transformer secondary.

A full wave rectifier consists of two half wave rectifiers, working independently and feeding a common load.

Therefore Primary TUF = 2 X TUF of HWR = 2 X 0.287

= 0.574

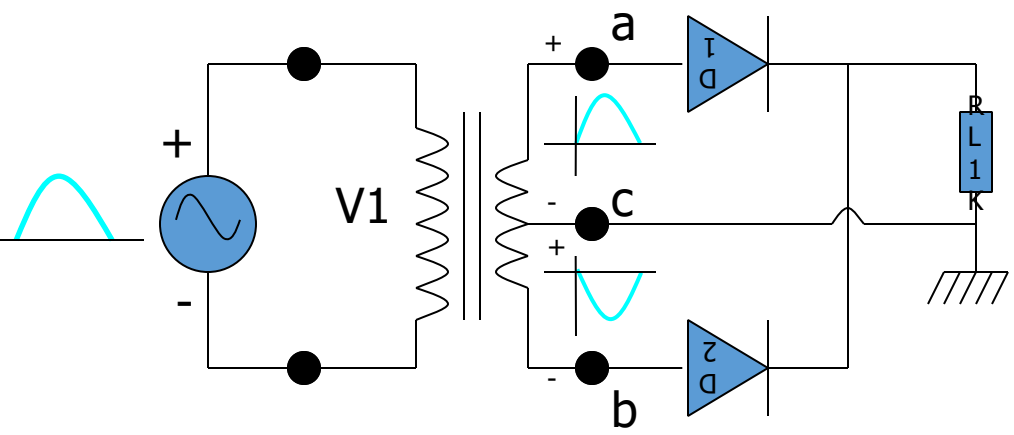
Therefore average TUF = [TUF (primary) + TUF (secondary)] / 2

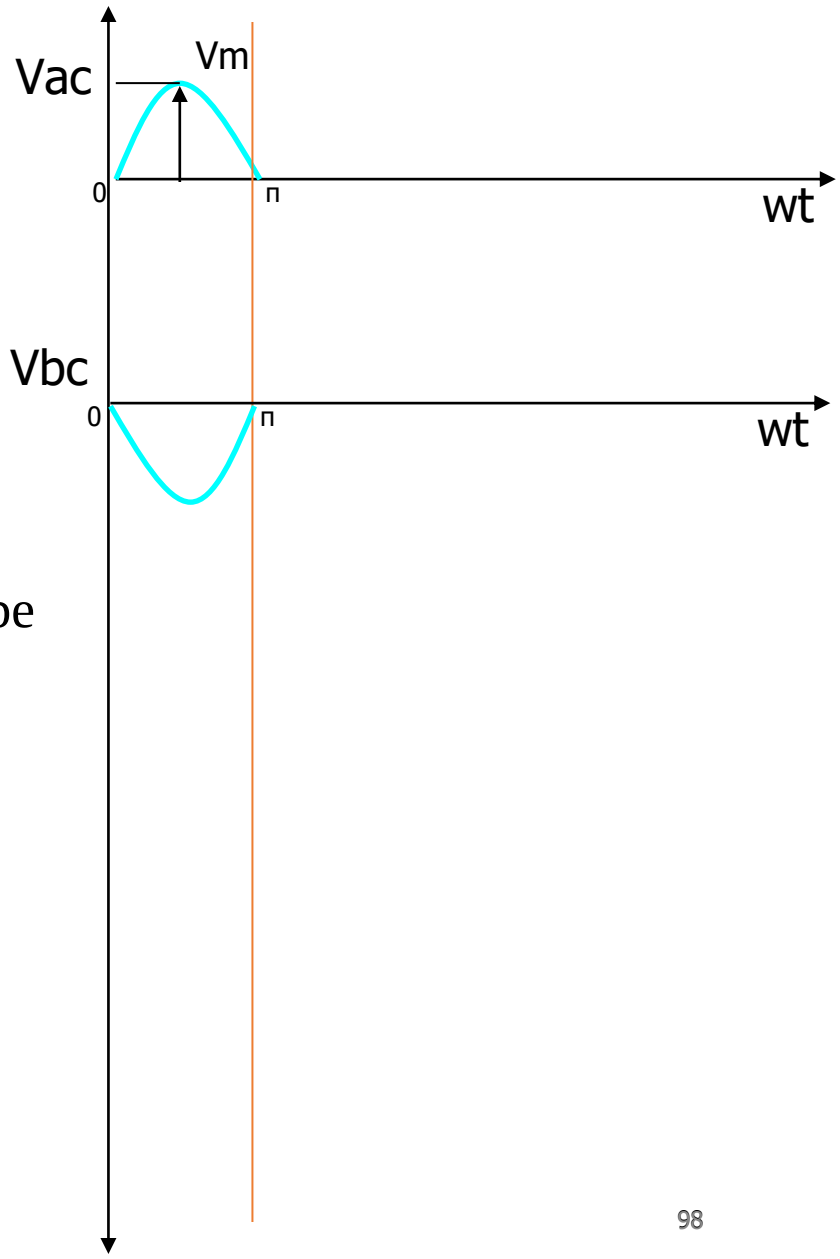
= (0.574 + 0.812) / 2

= 0.693 or 69.3 %

- This value of TUF is much higher than the TUF for HWR
- Hence transformer utilized is much more effectively by the FWR circuit.

Peak Inverse Voltage (PIV)



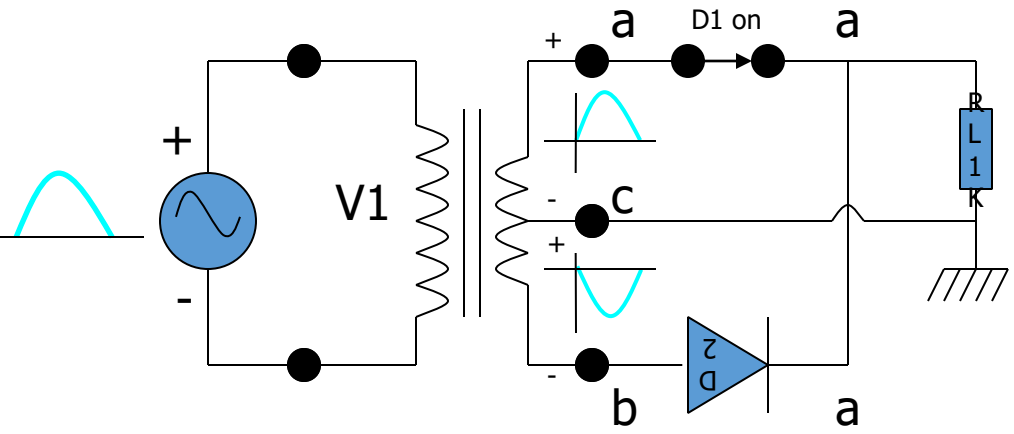


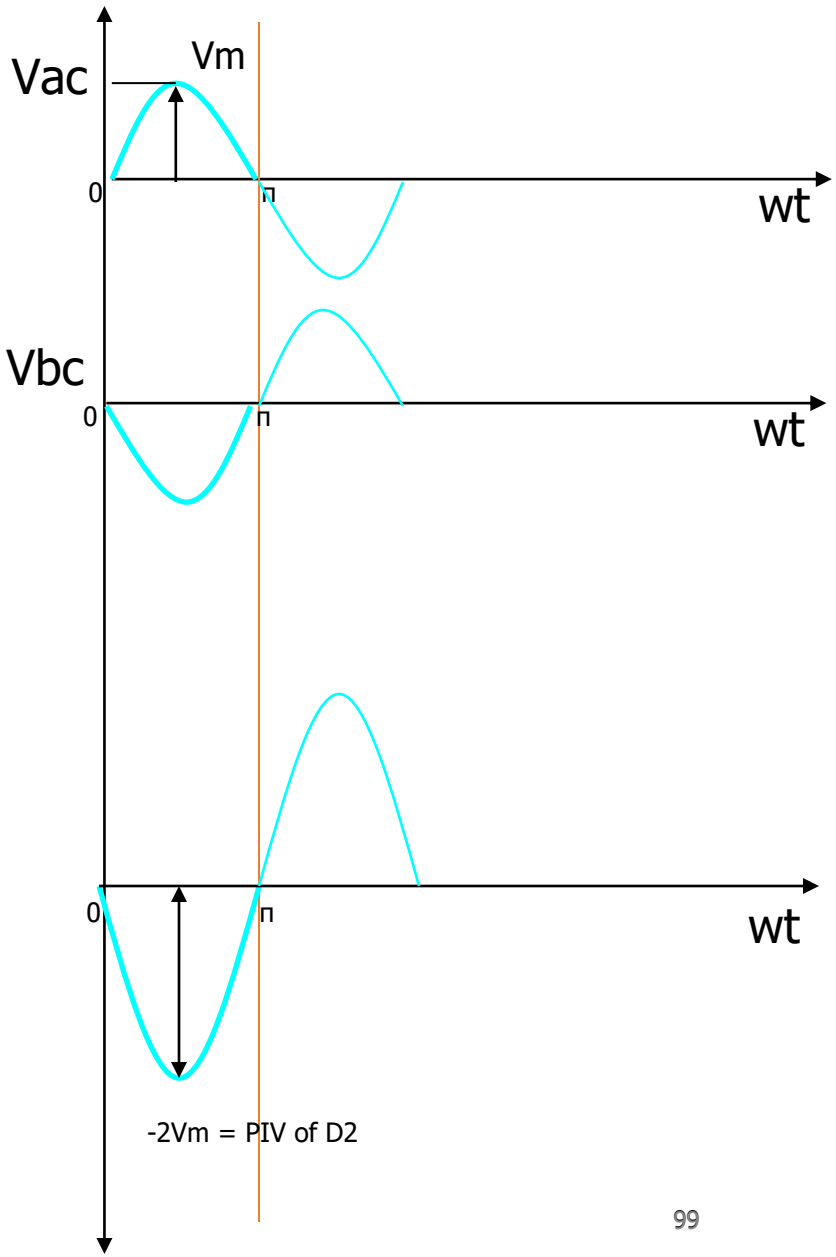
Diode D1 is conducting and it is assumed to be Equivalent to a closed switch.

Peak Inverse Voltage (PIV)

Figure shows that in the positive half cycle the instantaneous voltage across D2 is Vab

As shown in waveform the maximum negative Value of Vab is - Vm



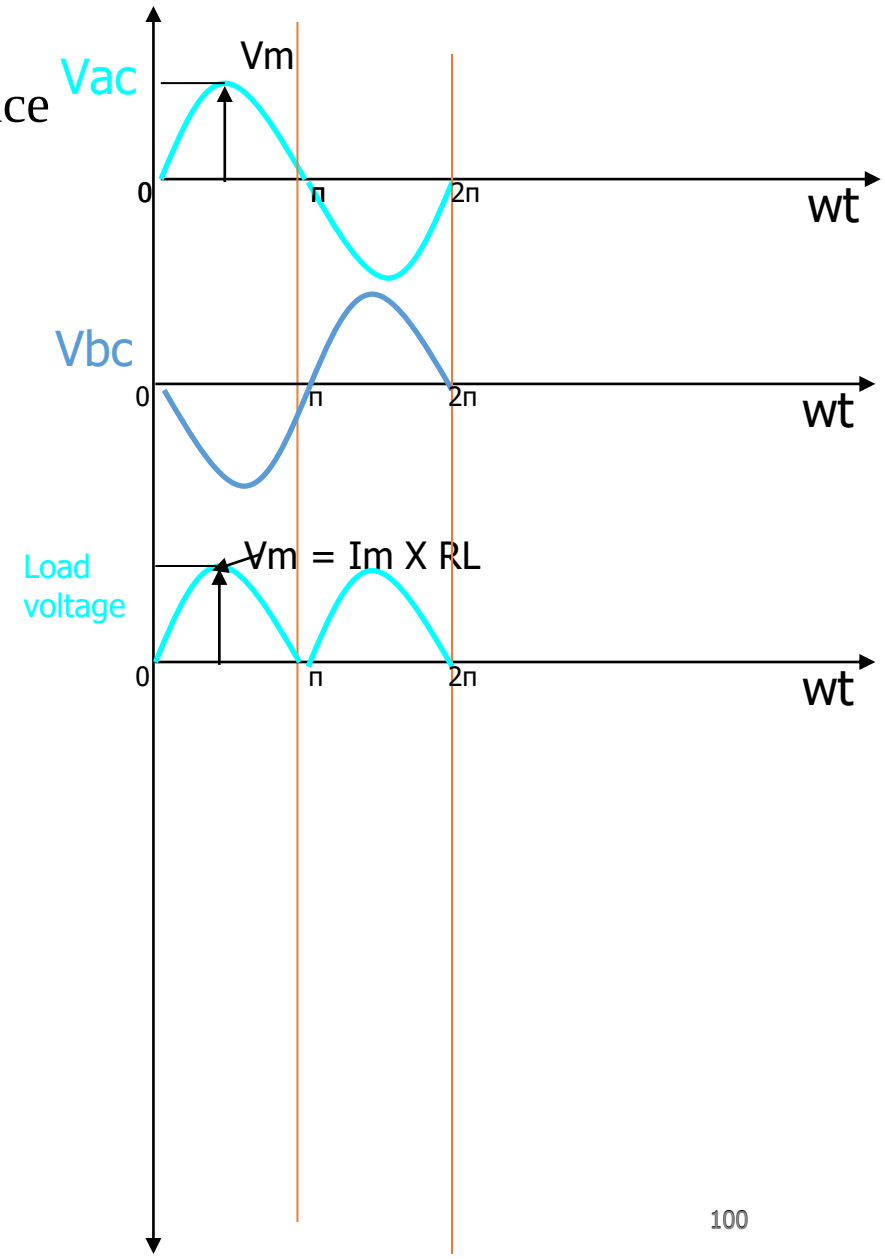


Therefore PIV = 2 Vm volts

Ripple Frequency Of Full Wave Rectifier

- The ripple frequency of FWR rectifier is twice Of the HWR is 100 Hz.

- This is the advantageous because the filter design becomes simpler as ripple frequency Is increased



Voltage Regulation

- Ideally the average rectifier output voltage should remain constant. But practically it varies with the changes in the load current
- Voltage regulation is defined as:

- Voltage regulation = $\frac{V_{NL} - V_{FL}}{V_{FL}}$ X 100%

But $V_{NL} = 2 V_M / \pi$ and

$V_{FL} = [2V_M / \pi] / [R_L / (R_S + R_F + R_L)]$

Voltage regulation = [(RS + RF) / RL] X 100

The voltage regulation is same as that of HWR

Performance Parameter Of A Full Wave Rectifier

Parameter	Value
Average DC load current (I_{LDC})	$2I_M/\pi$
Average DC load Voltage (V_{LDC})	$2V_M/\pi$
RMS value of load current (I_{LRMS})	$I_M/\sqrt{2}$
RMS value of load voltage (V_{LRMS})	$V_M/\sqrt{2}$
DC output power (P_{DC})	$(4I^2_m / \pi^2)$ X R_L
AC input power (P_{AC})	$(I^2_m / 2)$ X $(R_L + R_F + R_S)$
Maximum rectification efficiency (η_{max})	0.812 or 81.2 %
Ripple factor (RF)	0.48 or 48%
Ripple frequency	100 Hz
Peak inverse voltage (PIV)	2 Vm
Average transformer utilization factor (TUF)	0.693 or 69.3%

Advantage of Full Wave Rectifier

- The dc load voltage and current are more than half wave.
- Better rectification efficiency
- Low ripple factor as compared to HWR
- Better TUF

Disadvantage of Full Wave Rectifier

(i) It is difficult to locate the centre tap on the secondary winding.

(ii) The d.c. output is small as each diode utilises only one-half of the transformer secondary voltage.

(iii) The diodes used must have high peak inverse voltage.

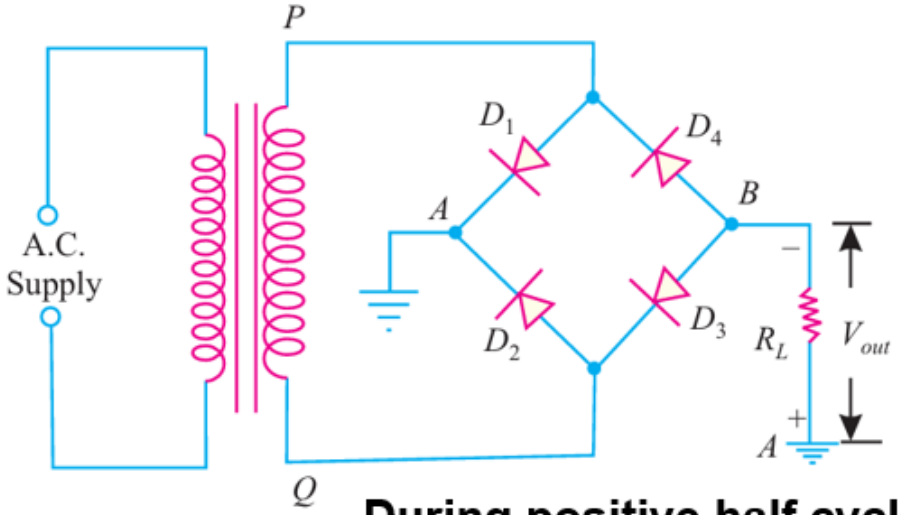
Comparison Of A Full Wave Rectifier And Half Wave Rectifier

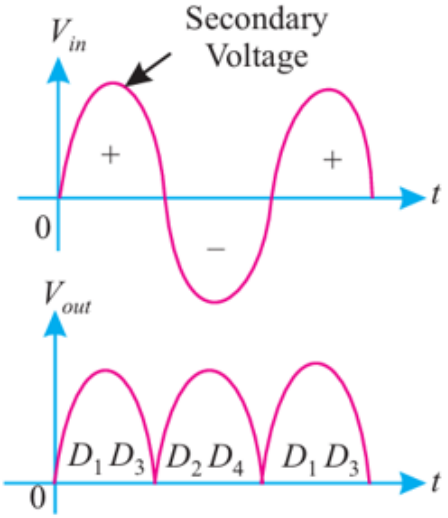
Parameter	Value HWR	FWR
Average DC load current (I_{LDC})	I_M/π	$2I_M/\pi$
Average DC load Voltage (V_{LDC})	V_M/π	$2V_M/\pi$
RMS value of load current (I_{LRMS})	$I_M/2$	$I_M/\sqrt{2}$
RMS value of load voltage (V_{LRMS})	$V_M/2$	$V_M/\sqrt{2}$
DC output power (P_{DC})	$(I^2_m / \pi^2) X R_L$	$(4I^2_m / \pi^2) X R_L$
AC input power (P_{AC})	$(I^2_m / 4) X (R_L + R_F + R_S)$	$(I^2_m / 2) X (R_L + R_F + R_S)$
Maximum rectification efficiency (η_{max})	0.4 or 40 %	0.812 or 81.2 %
Ripple factor (RF)	1.21or 121% very high	0.48 or 48%
Ripple frequency	50 Hz	100 Hz
Peak inverse voltage (PIV)	Vm	2 Vm
Average transformer utilization factor (TUF)	0.287or 28.7%	0.693 or 69.3%

Bridge Rectifier

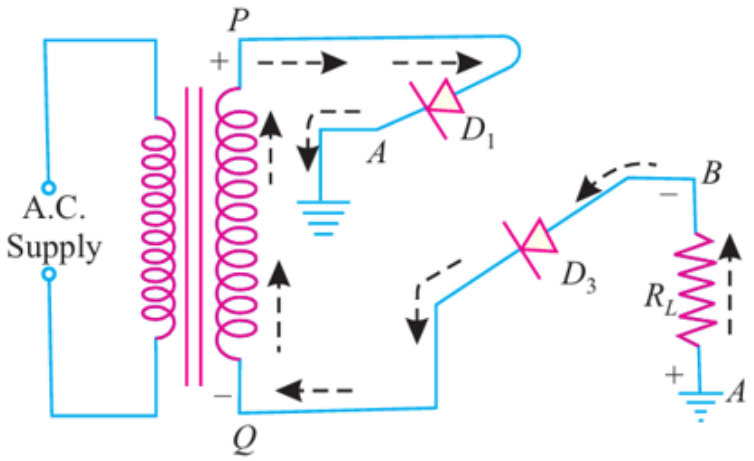
- Disadvantages of the full wave rectifier such as high PIV and compulsory use of center tapped transformer are overcome in the bridge rectifier.

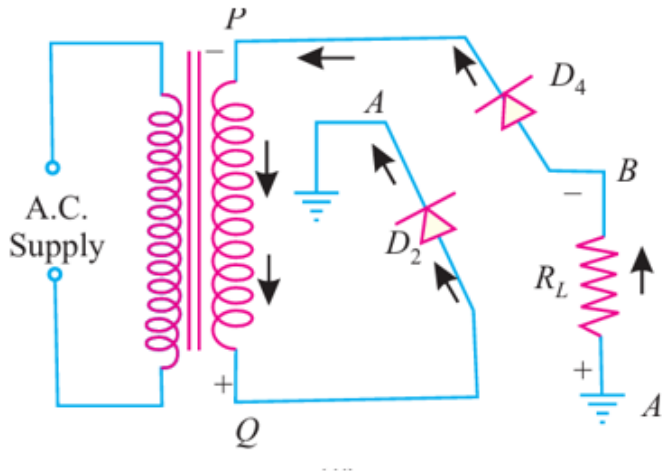
- The circuit configuration of a bridge rectifier, it consixt of four diodes connected to form a bridge





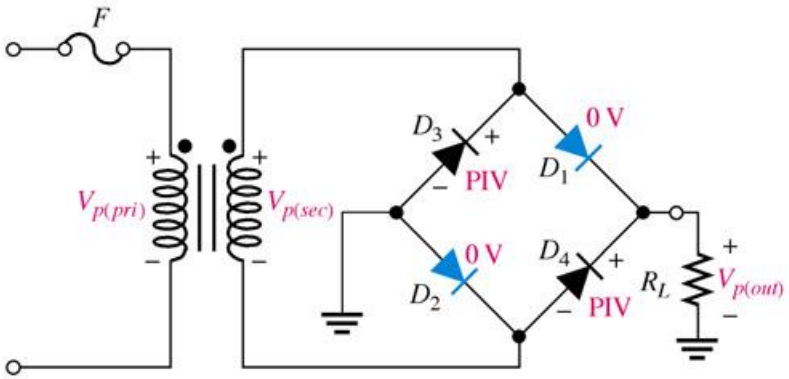
During positive half cycle



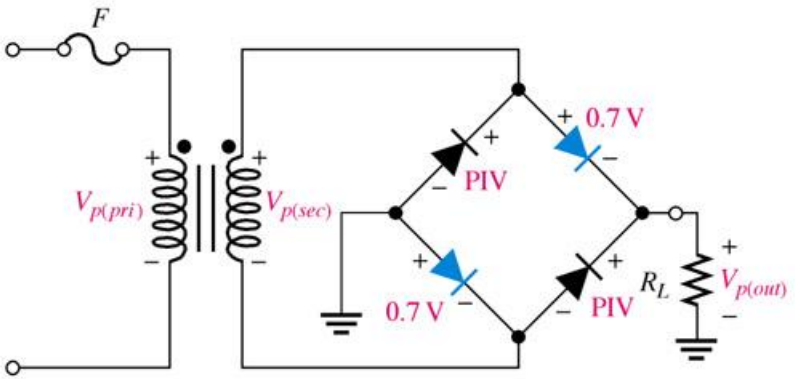


During negative half cycle

Peak inverse voltages across diodes D_3 and D_4 in a bridge rectifier during the positive half-cycle of the secondary voltage.



(a) For the ideal diode model (forward-biased diodes D_1 and D_2 are shown in blue). PIV = $V_{p(out)}$



(b) For the practical diode model (forward-biased diodes D_1 and D_2 are shown in blue). PIV = $V_{p(out)}$ + 0.7 V

Comparison Of A Full Wave Rectifier And Half Wave Rectifier

Parameter	Value HWR	FWR and Bridge Rectifier
Average DC load current (I_{LDC})	I_M/π	$2I_M/\pi$
Average DC load Voltage (V_{LDC})	V_M/π	$2V_M/\pi$
RMS value of load current (I_{LRMS})	$I_M/2$	$I_M/\sqrt{2}$
RMS value of load voltage (V_{LRMS})	$V_M/2$	$V_M/\sqrt{2}$
DC output power (P_{DC})	$(I^2_m / \pi^2) X R_L$	$(4I^2_m / \pi^2) X R_L$
AC input power (P_{AC})	$(I^2_m / 4) X (R_L + R_F + R_S)$	$(I^2_m / 2) X (R_L + R_F + R_S)$
Maximum rectification efficiency (η_{max})	0.4 or 40 %	0.812 or 81.2 %
Ripple factor (RF)	1.21or 121% very high	0.48 or 48%
Ripple frequency	50 Hz	100 Hz
Peak inverse voltage (PIV)	Vm	2 Vm for Bridge Vm
Average transformer utilization factor (TUF)	0.287or 28.7%	0.693 or 69.3% For Bridge 81.2%